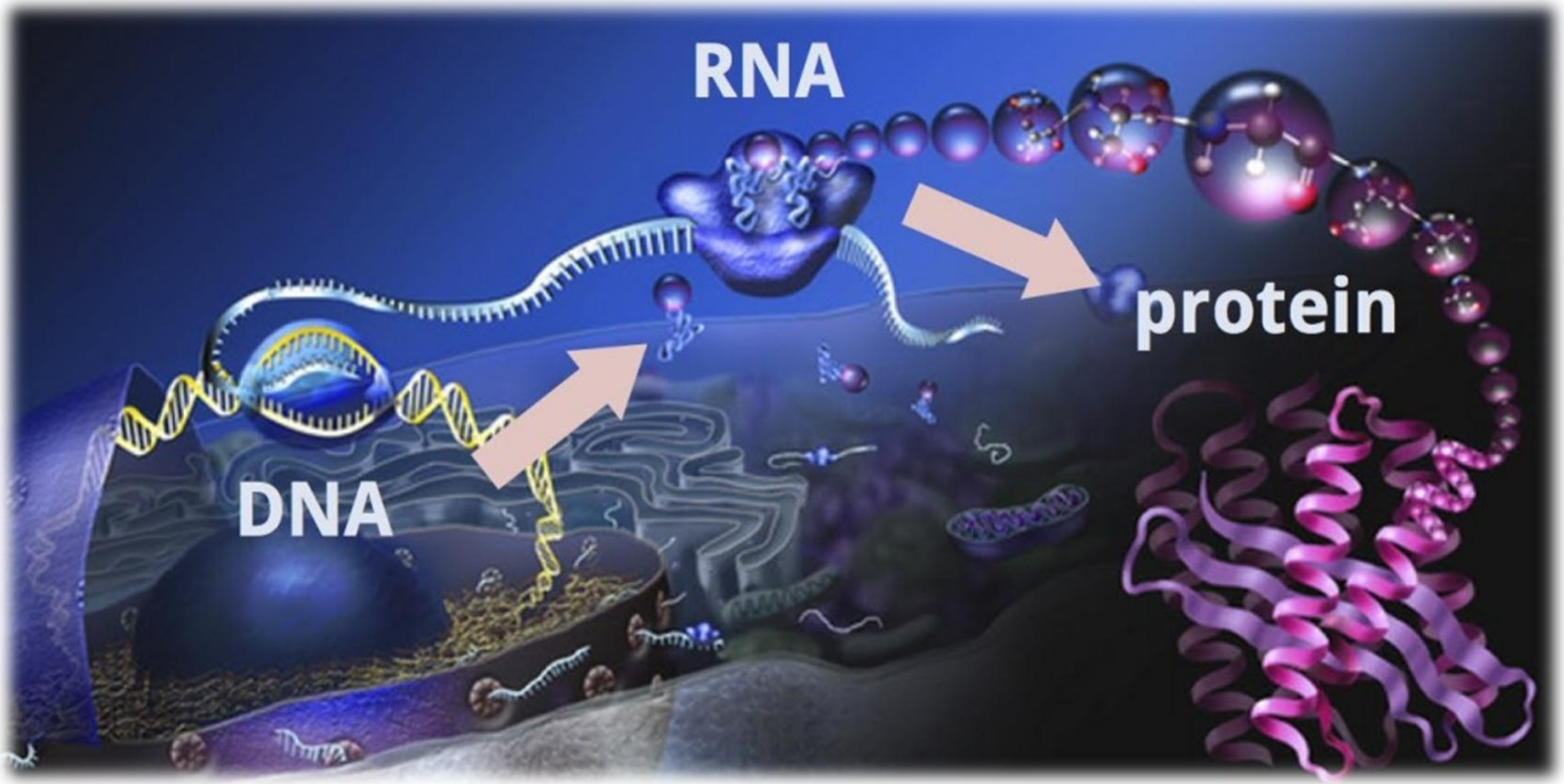
# ANALYSIS OF TRANSCRIPTOMES





***Rubén Sancho; Bruno Contreras-Moreira***


***16 December 2025 (6 hours)***
*9:00-10:00*
*10:00-11:00*
*11:30-13:30*
*15:00-17:00*

CIHEAM ZARAGOZA

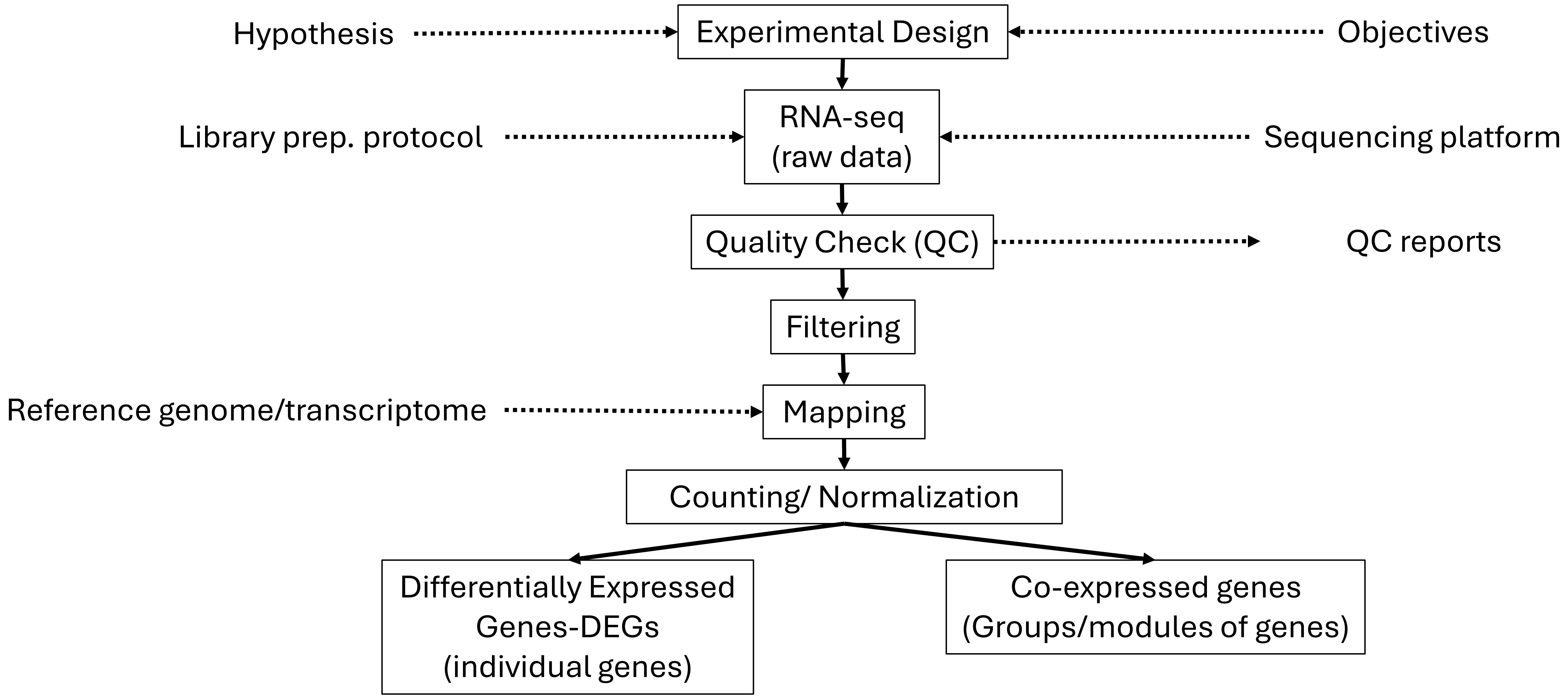
EEAD
Estación Experimental de Aula Dei

# Session Objectives

- ✓ Gain an overview of the complete RNA-seq analysis workflow.
- ✓ Understand differential gene expression (DEG) and co-expression network analyses.
- ✓ Understand the key concepts underlying these analyses.
- ✓ Become familiar with the bioinformatics tools and processes used in RNA-seq analysis.

# How to analyze RNA-seq data?

**From raw data to interpreting differential expression results, co-expression networks, and biological implications.**



# Experimental Design & RNA-seq

**Experimental design:** ✓ Tissues ✓ Samples ✓ Reps ✓ Treatments ✓ Sampling time

**Library type:** ✓ mRNA-seq (PolyA+) ✓ Total RNA-seq (no rRNA) ✓ Strand-specific RNA-seq ✓ Small RNA-seq ✓ scRNA-seq ✓ 3'-end RNA-seq

*The most widely used protocol involves the **TruSeq** (Illumina) kit, which assumes abundant starting material (0.1–1 μg total RNA) and maintains strand specificity*

**Sequencing (read) type:** ✓ Single-End ✓ Paired-end ✓ Short ✓ Long

**Goals:** ✓ Differentially Expressed Genes ✓ Co-expressed genes ✓ Isoform/Gene level …

**The choice of bioinformatics tools and commands depends on this information!!!**

## RNA-seq Library Types

| Type | Preparation | Captured RNA | Advantages | Disadvantages | Applications |
|---|---|---|---|---|---|
| **mRNA-seq (PolyA+)** | Selection of polyadenylated RNA using oligo(dT) beads | Mature mRNA (polyA+) | High specificity for **coding genes**; reproducible; high proportion of informative reads | Does not detect noncoding or non-polyA RNAs; unsuitable for degraded RNA | Differential gene expression; standard transcriptome profiling |
| **Total RNA-seq** | Ribosomal RNA removal (Ribo-Zero, NEBNext, etc.) | All RNAs except rRNA | Captures **coding and noncoding RNAs**; works with degraded RNA | Higher background noise; more expensive | Whole transcriptome profiling; lncRNA studies; FFPE* samples |
| **Strand-specific RNA-seq** | Incorporation of strand-specific markers during cDNA synthesis | Same as mRNA-seq or Total RNA-seq, but retains strand orientation | **Distinguishes sense/antisense** transcripts; improves isoform annotation | Slightly higher cost and complexity | Antisense RNA, isoform detection, transcriptional regulation |
| **Small RNA-seq** | Size selection (18–30 nt fragments) | miRNA, siRNA, piRNA, small snRNA** | Enables **post-transcriptional** regulation studies | Does not detect mRNA or long transcripts | miRNA profiling, piRNA discovery, small RNA pathways |
| **3’-end RNA-seq (TagSeq, QuantSeq)** | Captures only the 3’ ends of transcripts using short oligo(dT) priming | 3’ end of mRNA | Cost-effective; simple; correlates well with **gene expression** | No isoform or splicing information | Rapid expression quantification; high-throughput screening |
| **Single-cell RNA-seq (scRNA-seq)** | Cell barcoding using droplets or nanowells | Total RNA from individual cells | Single-cell resolution; **reveals cellular heterogeneity** | Expensive; lower coverage per cell; complex analysis | Developmental biology; cancer heterogeneity; immune profiling |
| **Exome-capture RNA-seq** | Hybridization with probes targeting known exons | Exonic coding regions | Focused on **genes of interest**; reduces background | Misses noncoding RNAs and unannotated isoforms | Targeted studies; validation of candidate genes |

# Quality Check (QC)

**In the Lab:** concentrations, absorbances and integrity (RIN- RNA Integrity Number 1-10)

| | Optimal Value | Minimum | Comments |
|---|---|---|---|
| **A260/A280** | 1.8–2.1 | ≥ 1.7 | Low values indicate protein or phenol contamination. |
| **A260/A230** | >2.0 | ≥ 1.8 | Low values suggest salt or reagent contamination. |
| **Concentration** | 100 ng – 2 µg total RNA | ≥ 50 ng | Depends on protocol; higher input improves library complexity. |
| **RIN (RNA Integrity Number)** | ≥ 8 | ≥ 6 (for rRNA-depleted RNA-seq) | Below 6, use specialized protocols (e.g., 3’-end RNA-seq). |

**In silico (check QC reports):** Per base sequence quality; Per tile sequence quality; Per sequence quality scores; Per base sequence content; Per sequence GC content; Per base N content; Sequence Length Distribution; Sequence Duplication Levels; Overrepresented sequences; Adapter Content

# Filtering

***Remove adapters and low-quality sequences and/or nucleotides***

## Trimmomatic software

•ILLUMINACLIP: Cut adapter and other Illumina-specific sequences from the read.

•SLIDINGWINDOW: Perform a sliding window trimming, cutting once the average quality within the window falls below a threshold.

•LEADING: Cut bases off the start of a read, if below a threshold quality

•TRAILING: Cut bases off the end of a read, if below a threshold quality

•CROP: Cut the read to a specified length

•HEADCROP: Cut the specified number of bases from the start of the read

•MINLEN: Drop the read if it is below a specified length

•TOPHRED33: Convert quality scores to Phred-33

•TOPHRED64: Convert quality scores to Phred-64

# Mapping and Counting

Mapping or aligning the sequences against a reference genome/transcriptome is a key step in identifying which genes the sequenced RNA fragments originate from and thus quantifying their expression.

To select the alignment/mapping software we must take into account:

- ✓ What reference is available, genome or transcriptome?
- ✓ What downstream analyses will be performed: differential expression, co-expression networks, ...?

| Software | Alignment type | Reference | Speed | Accuracy | Splicing detection | Direct quantification | Output | Key notes |
|---|---|---|---|---|---|---|---|---|
| **STAR** | *Full alignment (sequence-based)* | **Genome** | Very high | Very high | Yes | **No** (use featureCounts or RSEM) | SAM/BAM files | Gold standard in large projects (e.g., ENCODE, GTEx); excellent for novel genes or isoforms |
| **HISAT2** | *Full alignment (hierarchical indexing)* | **Genome** | High | High | Yes | **No** (use StringTie or featureCounts) | SAM/BAM files | Fast and memory-efficient; successor to TopHat2 |
| **Salmon** | *Quasi-mapping* | **Transcriptome** | Very high | High | Not directly (depends on annotation) | **Yes** | **Expression tables** (TPM, FPKM, counts) | Very fast; ideal for direct quantification without BAM files |
| **Kallisto** | *Pseudoalignment* | **Transcriptome** | Extremely high | High | No | **Yes** | **Expression tables** (TPM, counts) | Ultra-fast; great for exploratory or large-cohort analyses |

# Counting, Normalization and Differential Expression analysis

## Gene Expression Measurement

**Raw counts:** Number of sequencing reads mapped to each gene.
Cannot be directly compared between genes or samples:

- Longer genes = more sequenced reads
- Samples with deeper sequencing = more total sequenced reads

***Normalize for sequencing depth and gene length***

**RPKM:** Reads Per Kilobase of exon per Million reads mapped. ***Single-end*** (SE) RNA-seq.
**FPKM:** Analogous to RPKM and is used specifically in ***paired-end*** (PE) RNA-seq experiments.

$$RPKM_i \text{ or } FPKM_i = \frac{q_i}{\frac{l_i}{10^3} * \frac{\sum_j q_j}{10^6}} = \frac{q_i}{l_i * \sum_j q_j} * 10^9$$

$q_i$ → raw read or fragment counts (per feature and sample)

$l_i$ → feature (i.e., gene or transcript) length (per feature, bp)

$\sum_j q_j$ → total number of mapped reads or fragments (per sample)

**TPM:** Transcript Per Million.
The sum of all TPM values is the same in all samples. A TPM value represents a relative expression level that should be comparable between samples.

$$TPM_i = \frac{q_i / l_i}{\sum_j (q_j / l_j)} * 10^6$$

$q_i$ → reads mapped to transcript (per transcript and sample)

$l_i$ → transcript length (per transcript bp)

$\sum_j (q_j / l_j)$ → the sum of mapped reads to transcript normalized by transcript length

# Tools for Counting/Normalization and Differential Expression analysis

| Tool | Type | Input | Output | Level | Use | Advantages | Disadvantages |
|---|---|---|---|---|---|---|---|
| **featureCounts** | Counting | Aligned reads (BAM/SAM) | Raw counts | Gene | Read summarization | Fast, memory-efficient | No isoform quantification |
| **RSEM** | Quantification | Aligned reads (or raw FASTQ) | TPM, FPKM, counts | Gene & transcript | Quantifying expression | Handles multi-mapping reads, isoform-aware | Slow, computationally demanding |
| **StringTie** | Assembly + Quantification | Aligned reads | Assembled transcripts + FPKM | Transcript | Transcript reconstruction | Detect novel isoforms | Less accurate with low coverage |
| **DESeq2** | Differential expression | Raw counts | Fold change, p-values | Gene | Expression comparison | Statistically robust, well-documented | Requires raw counts, not TPM |
| **Sleuth** | Differential expression | TPMs (from Kallisto) | Fold change, q-values | Transcript | Fast DE analysis | Very fast, interactive, works with Kallisto | Kallisto input |

# Differentially Expressed Genes (DEGs) using Sleuth

*Genes whose expression levels significantly increase or decrease between different conditions or groups*

Different tissues (leaves, seeds, flower parts, roots, …), different growth stages, treatments/doses (e.g. nitrogen, …), abiotic stresses (heat, cold, drought, waterlogging, nutrient deficiency, …), or biotic stresses (diseases, …).

**`Sleuth` is used to explore gene expression results from `Kallisto` and to perform differential expression analysis.**

**`Sleuth` is implemented in an interactive shiny app that utilizes `Kallisto` quantifications and bootstraps for fast and accurate analysis of data from RNA-seq experiments.**

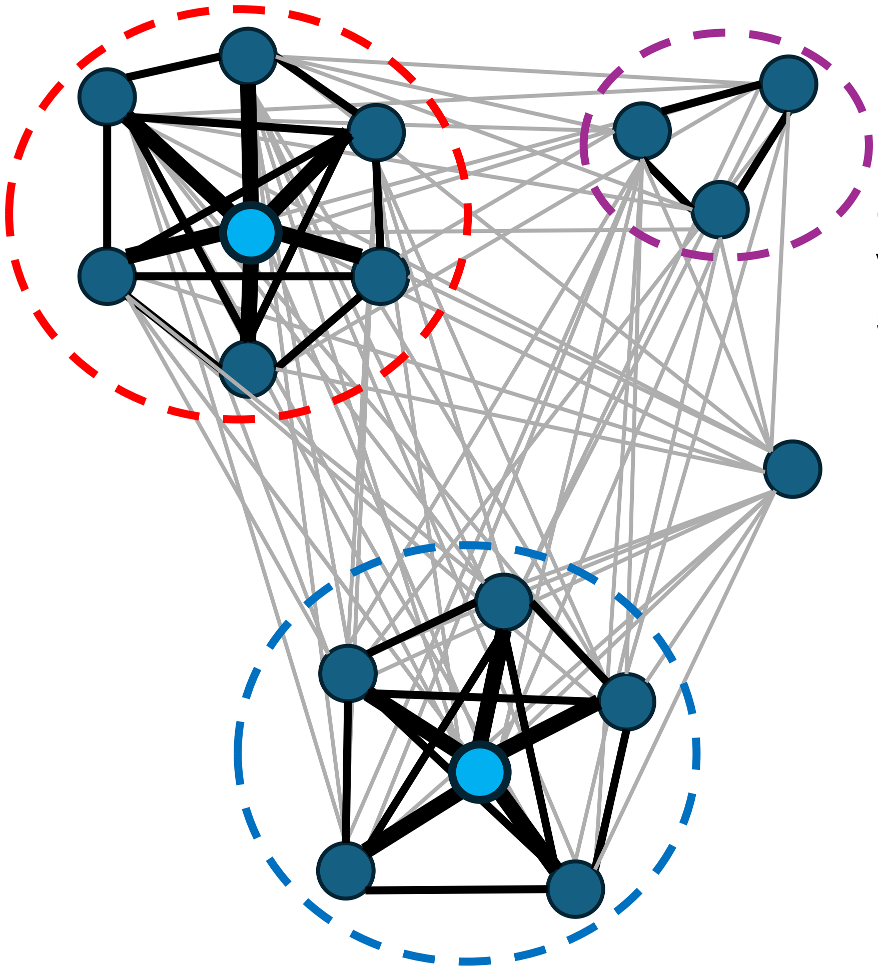
**`Sleuth` uses the bootstraps to ascertain and correct for technical variation in experiments.**

# Gene Co-expression Networks

Graphical representations that show how genes are related based on the **similarity of their expression patterns.**

**Node = Gene** — **Edge = Connection (correlated expression levels)**

***Premise:*** *Genes with similar expression patterns (co-expressed) are functionally associated*



**Weighted**

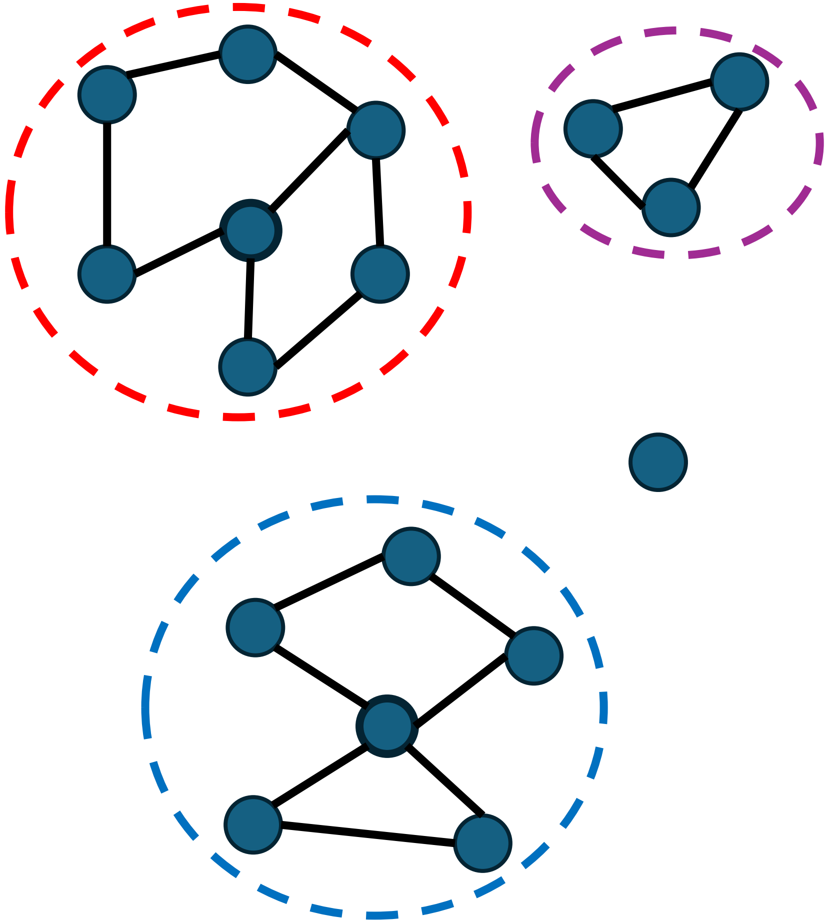
Connection strength varies (from 0 to 1)

Soft-threshold

**Unweighted**

Connected (1) or not connected (0)

Hard-threshold



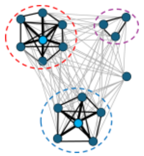
Hub Gene/Isoform (potential driver genes)

Gene/Isoform

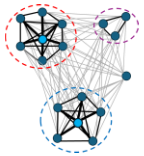
Weak connections

Strong connections

Very Strong connections



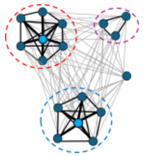
**WGCNA**
**Weighted Gene Co-expression Network Analysis**



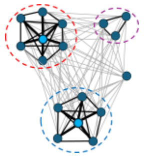
## Glossary of WGCNA Terminology

**1. Data set**

- ✓ Expression matrix (normalized expression values) with rows as genes/isoforms and columns as samples.
- ✓ Not filtering genes by differential expression.
- ✓ It is recommended that the analysis be performed with a larger number of samples, at least 20.
- ✓ Clean the data by removing obvious outlier samples as well as genes and samples with excessive numbers of missing entries.



# WGCNA
# Weighted Gene Co-expression Network Analysis



## 2. Weighted Network construction

**Adjacency:** matrix n x n which encodes the network **connection strength between nodes *i* and *j***.
An intermediate quantity called the *co-expression* ***similarity*** $s_{ij}$ is first defined to calculate the adjacency matrix.
The default method defines the co-expression **similarity** $s_{ij}$ **as the absolute value of the correlation coefficient** (Pearson, biweight or the Spearman correlation) between the profiles of nodes *i* and *j*: $s_{ij} = |cor(x_i, x_j)|$.
The co-expression **similarity is transformed into the adjacency using a thresholding procedure**.

*Unweighted networks:* adjacency $a_{ij}$ between gene expression profiles $x_i$ and $x_j$ can be defined by **hard thresholding** the co-expression similarity $s_{ij}$ as

$$a_{ij} = \begin{cases} 1 & \text{if } s_{ij} \geq \tau; \\ 0 & \text{otherwise,} \end{cases}$$

where $\tau$ is the "hard" threshold parameter. Thus, two genes are linked ($a_{ij} = 1$) if the absolute correlation between their expression profiles exceeds the (hard) threshold $\tau$.

***Weighted networks:*** the adjacency takes on continuous values between 0 and 1 and can be defined by raising the co-expression **similarity to a power**: $a_{ij} = s_{ij}^{\beta}$, with $\beta \geq 1$. The weighted adjacency $a_{ij}$ between two genes is proportional to their similarity.

***How to choose the threshold parameters?*** Scale-free topology criterion.

# WGCNA
# Weighted Gene Co-expression Network Analysis

***How to choose the threshold parameters?*** Scale-free topology criterion.

A ***scale-free network*** is one where ***most nodes*** have ***few connections***, and a ***few nodes*** (hubs) have ***many connections***. Many real biological systems show this property.

WGCNA package tests a range of β values (e.g. 1 to 20), and calculates:

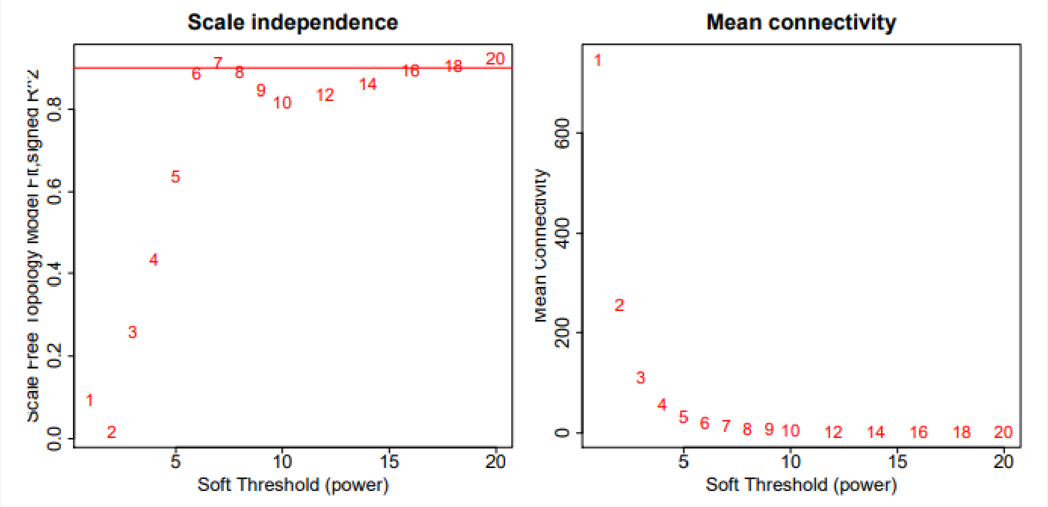
- ✓ The $R^2$ of the fit between node connectivity and the power-law distribution (how "scale-free" the network is)
- ✓ The mean connectivity (so the network isn't too sparse or too dense).

Then WGCNA plots these results:

- ✓ **$R^2$ vs. β** (to see where the scale-free fit becomes strong)
- ✓ **Mean connectivity vs. β** (to check network density).

The **best β** is usually the **smallest value** where:

- ✓ the **scale-free fit $R^2$** is **high (≥ 0.8–0.9)**, and
- ✓ the **mean connectivity** is still reasonable.

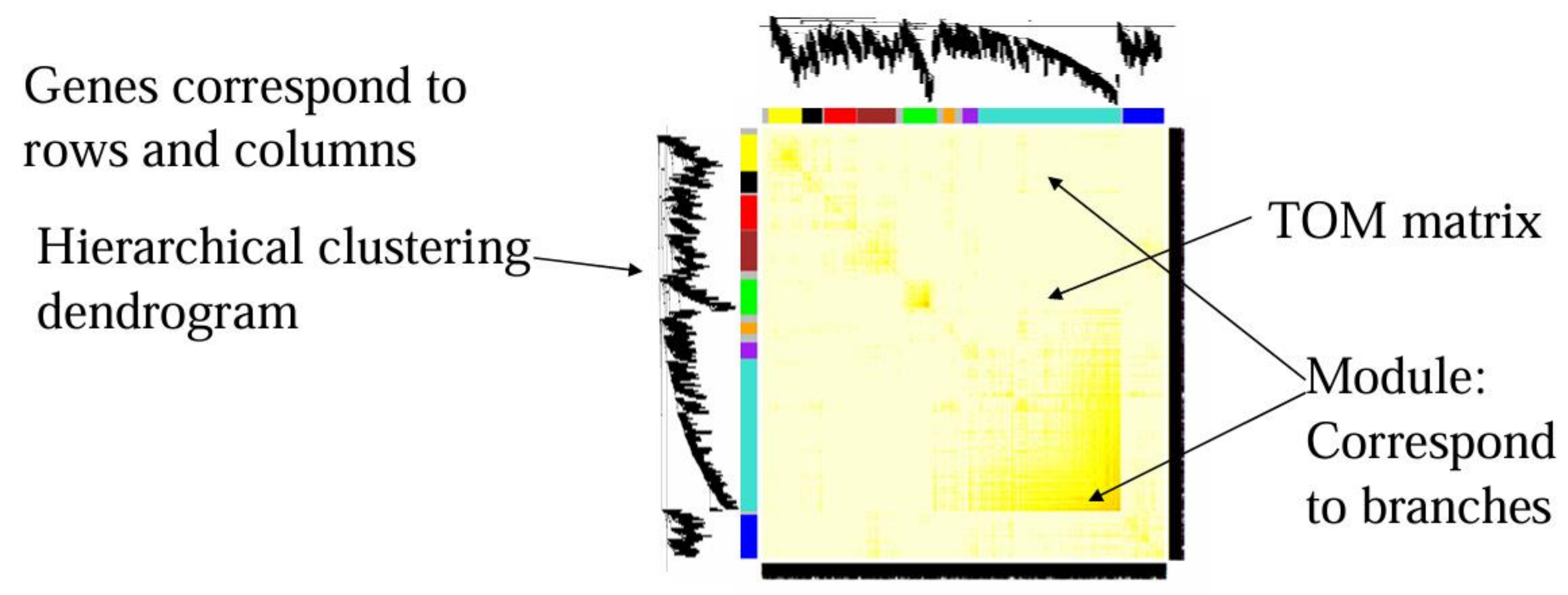


## 3. Module detection

Modules are clusters of highly interconnected genes and are defined by average linkage **hierarchical clustering** using as input the **topological overlap dissimilarity** (distance) measure (the higher the value, the more different the connection patterns of two genes). $DistTOM_{ij} = 1 - TOM_{ij}$

**Topological Overlap Matrix (TOM)** is a measure of how similar two genes are in terms of their network connections, considering not only the direct correlation between two genes but also **how many neighbors they share** (e.g. If gene A and gene B are both connected to many of the same genes, their TOM will be high).

$$TOM_{ij} = \frac{\sum_u a_{iu} a_{uj} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}}$$




A network heatmap plot (interconnectivity plot) of a gene network together with the corresponding hierarchical clustering dendrograms and the resulting modules

Source: An Overview of Weighted Gene Co-Expression Network Analysis by Steve Horvath

**WGCNA**

**Weighted Gene Co-expression Network Analysis**

**4.1. Gene significance (GS)**

- ✓ It used to incorporate external information into the co-expression network.
- ✓ The higher the absolute value of $GS_i$ , the more biologically significant is the *i-th* gene.
- ✓ A gene significance measure could also be defined by minus log of a p-value.

**4.2. Module significance**

The average absolute gene significance measure for all genes in a given module.

**5. Other key parameters**

**Connectivity:** it measures how correlated a gene is with all other network genes.

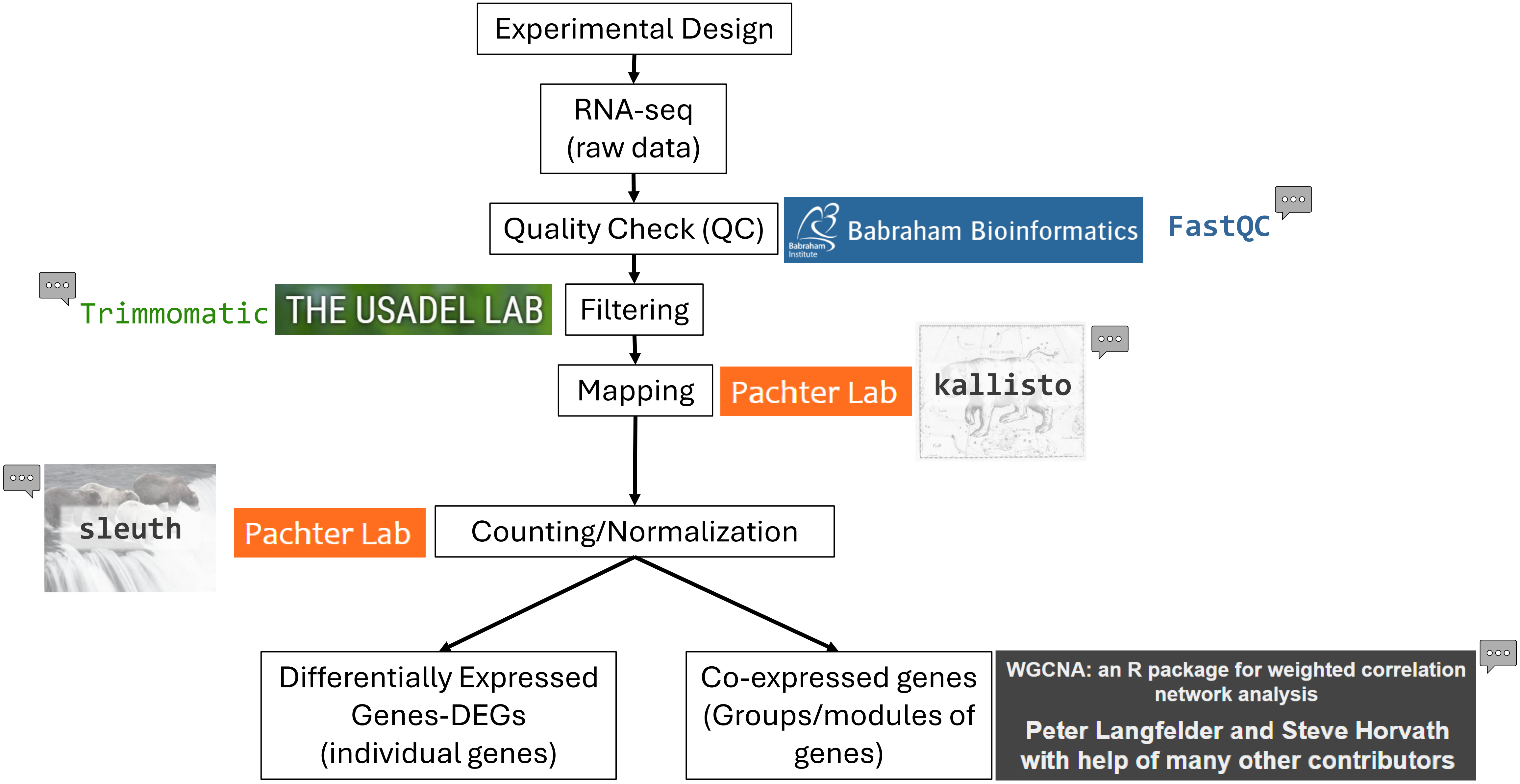
**Intramodular connectivity ($k_{IM}$):** it measures how connected, or co-expressed, a given gene is with respect to the genes of a particular module.

**Module eigengene E:** The first principal component of a given module (such as a representative of the gene expression profiles in a module).

**Eigengene significance:** The correlation coefficient between the module eigengene and a trait.

**Module Membership (MM)** (also known as eigengene-based connectivity $k_{ME}$)**:** For each gene, a measure of module membership by correlating its gene expression profile with the module eigengene of a given module.

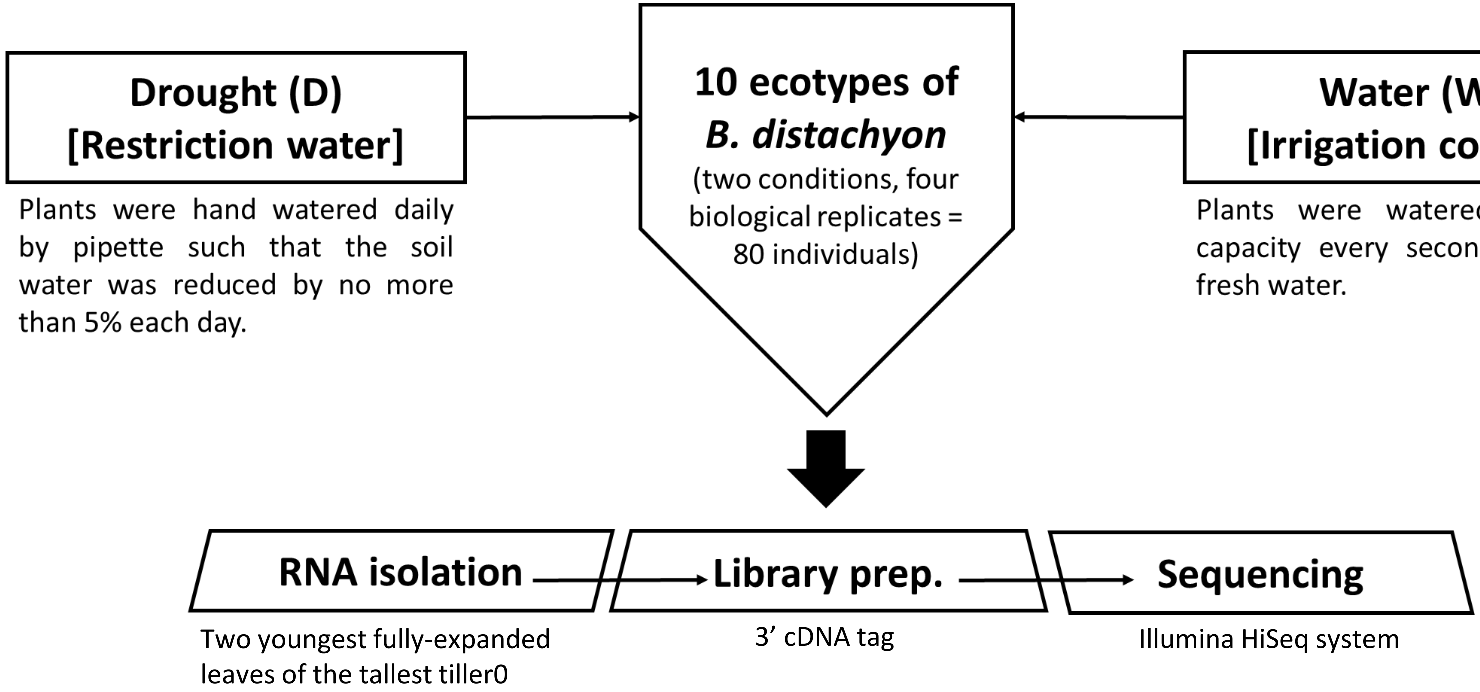
# Practical Exercise


Experimental Design
RNA-seq (raw data)
Quality Check (QC)
Babraham Bioinformatics
FastQC
Trimmomatic
THE USADEL LAB
Filtering
Mapping
Pachter Lab
kallisto
sleuth
Pachter Lab
Counting/Normalization
Differentially Expressed Genes-DEGs (individual genes)
Co-expressed genes (Groups/modules of genes)
WGCNA: an R package for weighted correlation network analysis
Peter Langfelder and Steve Horvath with help of many other contributors

# Experimental Design

**Hypothesis:** There are genes with similar expression profiles involved in the response to water stress.

**Objective:** To detect a set of co-expressed genes involved in the response to water stress and the hub genes with a key role in the regulation of the module.




Des Marais et al (2017)

Sancho et al. (2022)

# Readme (Step-by-step)

- OPEN the "**Readme_Docker_Transcriptomic_class.txt**" file
  - ✓ OPEN Ubuntu (WSL)
  - ✓ In your home, create the folder "`transcriptome_class`" and extend permissions
    - `mkdir transcriptome_class`
    - `chmod 777 transcriptome_class`
  - ✓ Docker environment: https://github.com/eead-csic-compbio/bioinformatics/tree/main/docker
    - Run (in Ubuntu): `docker pull csicunam/bioinformatics_iamz:latest`
    - Bind local "`transcriptome_class`" folder to container folder "/data":

```
docker run -it -v $HOME/transcriptome_class:/data -v /tmp/.X11-
unix:/tmp/.X11-unix -e DISPLAY=$DISPLAY csicunam/bioinformatics_iamz:latest
```

  - ✓ From **Windows**, the "**transcriptome_class"** folder is located in:

\\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class

\\wsl.localhost\Ubuntu-22.04\home\rsancho\transcriptome_class (Rubén Sancho laptop)

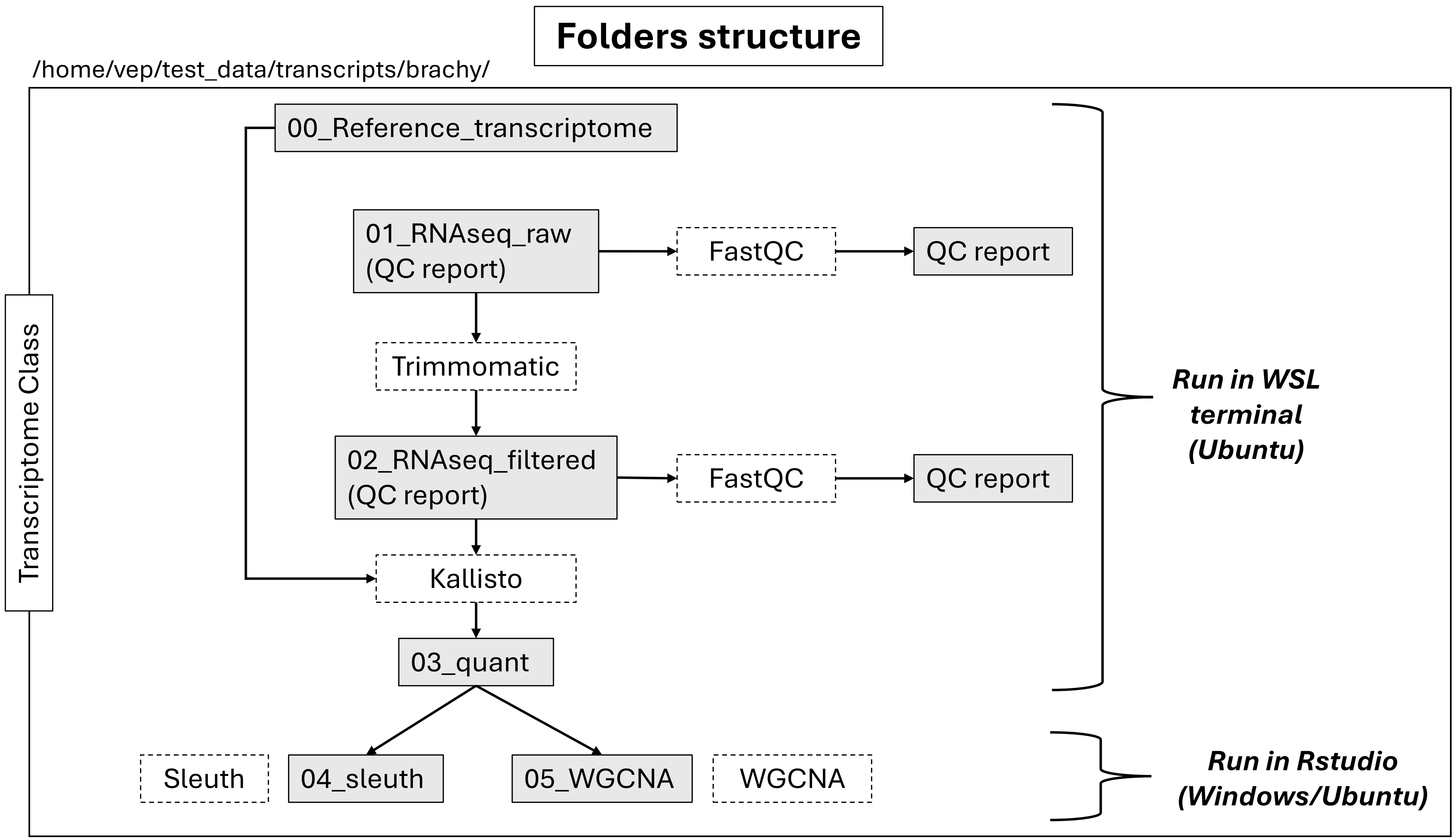
## Readme (Step-by-step)

- ✓ Move into /data: `cd /data`
  - Create symbolic link:

```
ln -s /home/vep/test_data/transcripts/brachy/ brachy
```

  - Create report folders:

```
mkdir report report_filtered
```


Folders structure
/home/vep/test_data/transcripts/brachy/
Transcriptome Class
00_Reference_transcriptome
01_RNAseq_raw
(QC report)
FastQC
QC report
Trimmomatic
02_RNAseq_filtered
(QC report)
FastQC
QC report
Kallisto
03_quant
Sleuth
04_sleuth
05_WGCNA
WGCNA
Run in WSL terminal (Ubuntu)
Run in Rstudio (Windows/Ubuntu)

# RNA-seq

RNA-Seq library protocol is the **3' cDNA tag library prep** for sequencing on the Illumina HiSeq platform adapted from Meyer et al. (2011).

Sequencing was carried out using an Illumina HiSeq2500 platform (100 bp Single-End, SE, sequencing).

# Quality Check (QC)

**Task 01: Generate QC report of ABR3_W_03 sample.**
***Review and interpret the results obtained***

**Software:** FastQC v0.12.1.

**Command:** `fastqc input_fastqc -o output_folder`

**Tips:**

- ✓ Remember to create the output folder.
- ✓ Check that the paths of the software, input files, and output folder are correct.

*Answer*

# Quality Check (QC)

```
fastqc 01_RNAseq_raw/ABR3_W_03.fastq.gz \
-o 01_RNAseq_raw/reports/
```

Open the report: \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\report\ABR3_D_03_fastqc.html

***Answer***

# Quality Check (QC) with FastQC

## FastQC Report

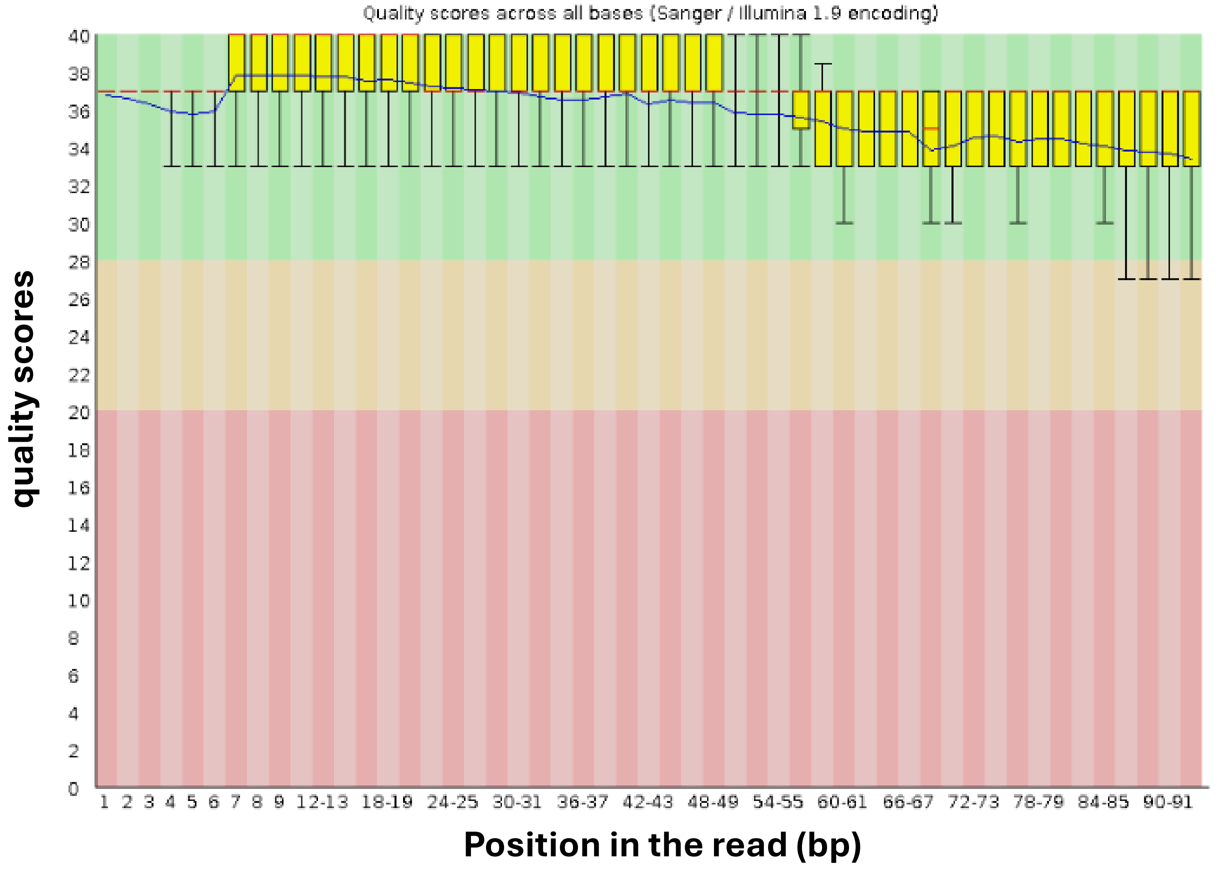
### Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

### Basic Statistics

| Measure | Value |
|---|---|
| Filename | ABR3_W_03.fastq.gz |
| File type | Conventional base calls |
| Encoding | Sanger / Illumina 1.9 |
| Total Sequences | 2705509 |
| Total Bases | 248.8 Mbp |
| Sequences flagged as poor quality | 0 |
| Sequence length | 70-93 |
| %GC | 45 |

### Per base sequence quality

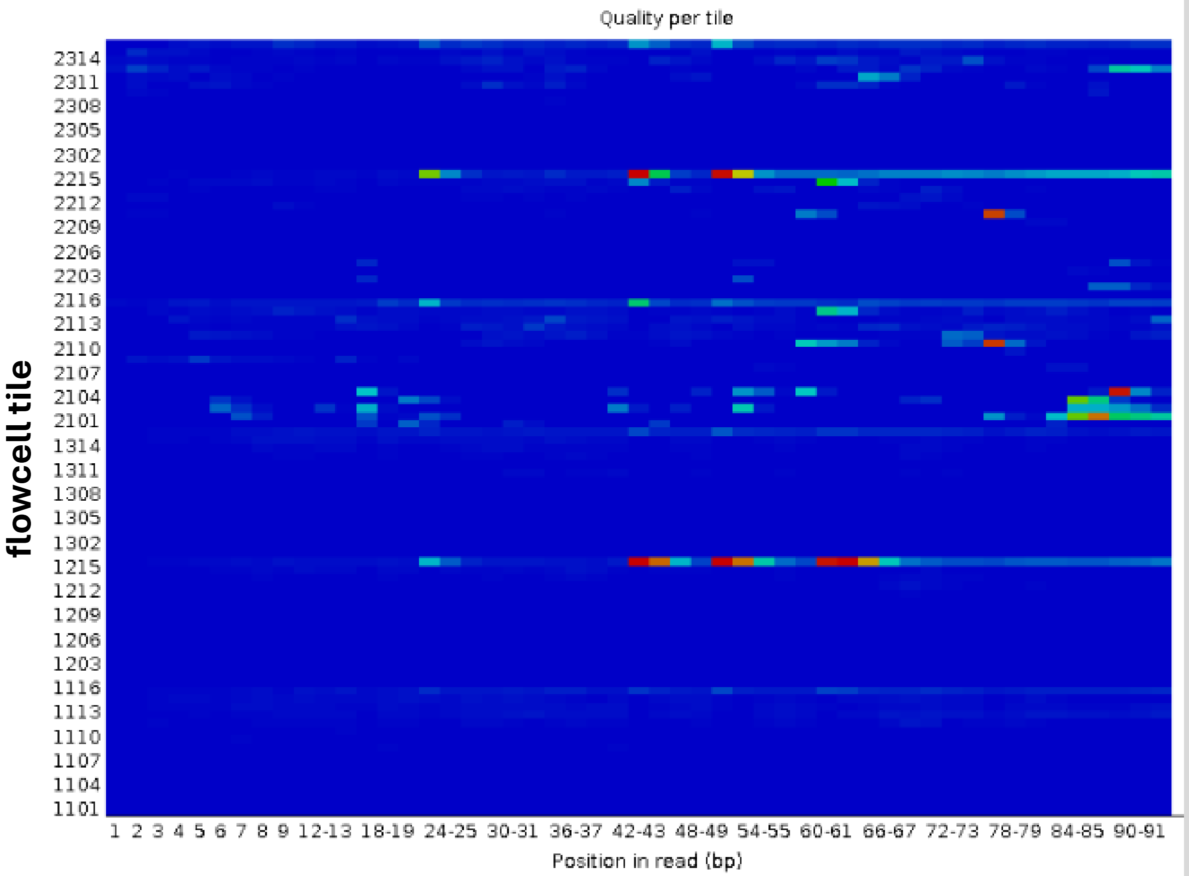



The range of quality scores across all bases at each position in the Fastq file

**Answer**

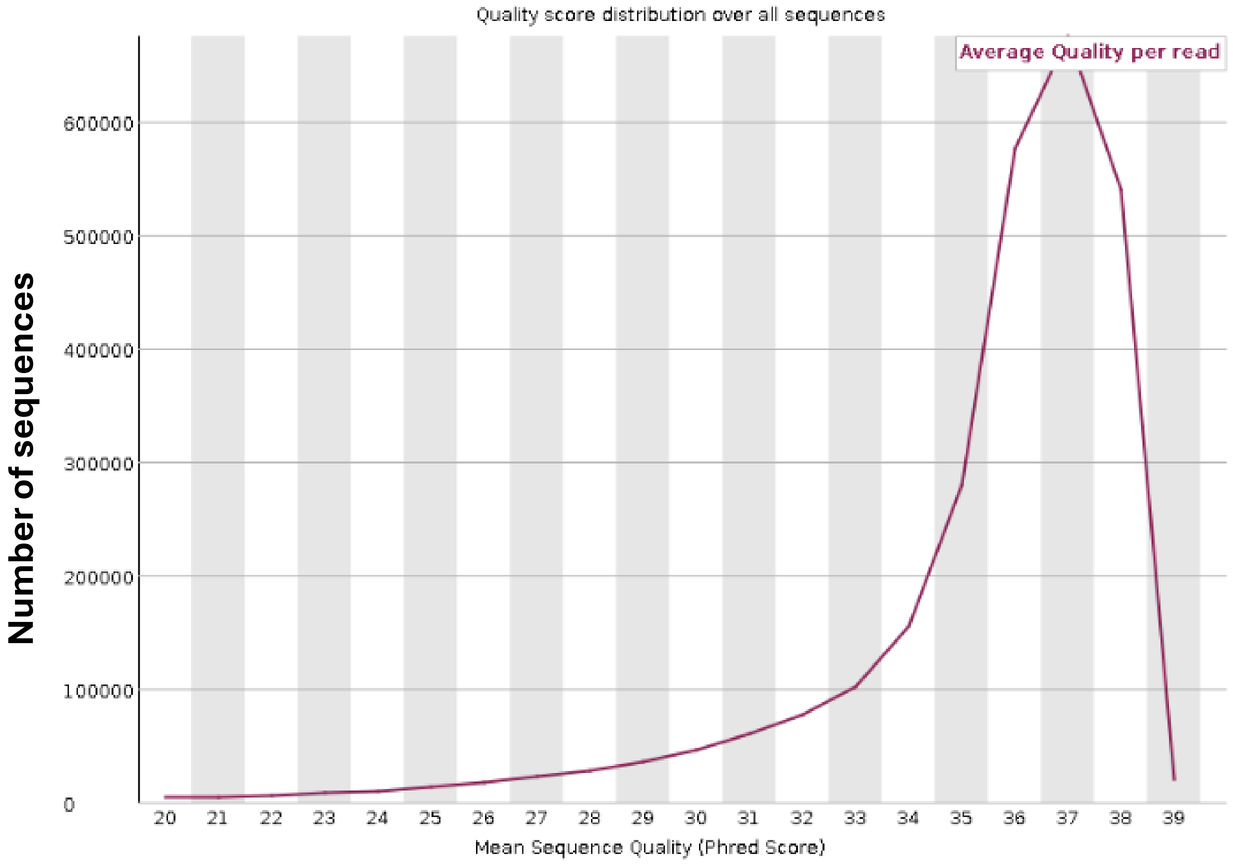
# Quality Check (QC) with FastQC

## Per tile sequence quality




The plot shows the deviation from the average quality for each flowcell tile. Hotter colours indicate that a tile had worse qualities than other tiles for that base. Bubbles, smudges or debris in the flowcell lane.

## Per sequence quality scores

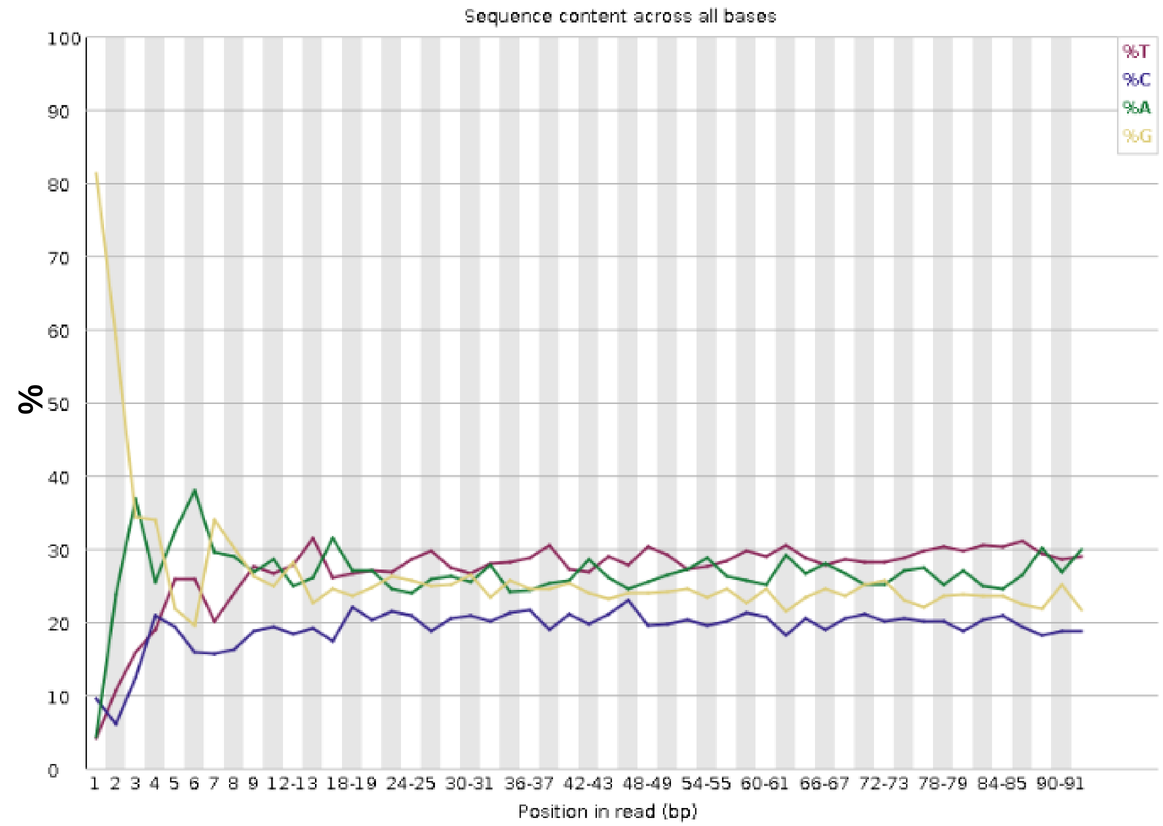



It allows you to see if a subset of your sequences have universally low-quality values.

**Answer**

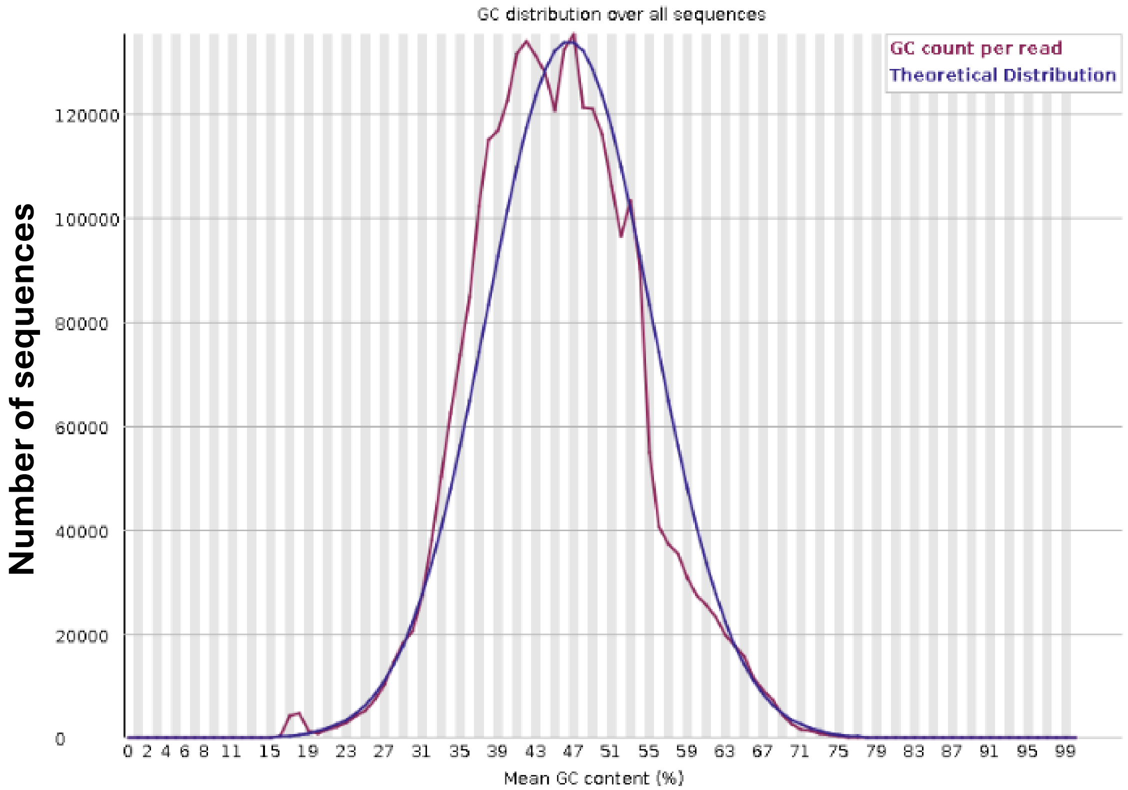
# Quality Check (QC) with FastQC

## Per base sequence content




The proportion of each base position in a file for which each of the four normal DNA bases has been called. **In a random library** you would expect that there would be little to no difference between the different bases (A-T and C-G) of a sequence run. ***IMPORTANT***: *Some types of library will always produce biased sequence composition, normally at the start of the read. Libraries produced by priming using random hexamers (including nearly all RNA-Seq libraries) inherits an intrinsic bias in the positions at which reads start. This is not a problem which can be fixed by processing, and it doesn't seem to adversely affect the ability to measure expression.*

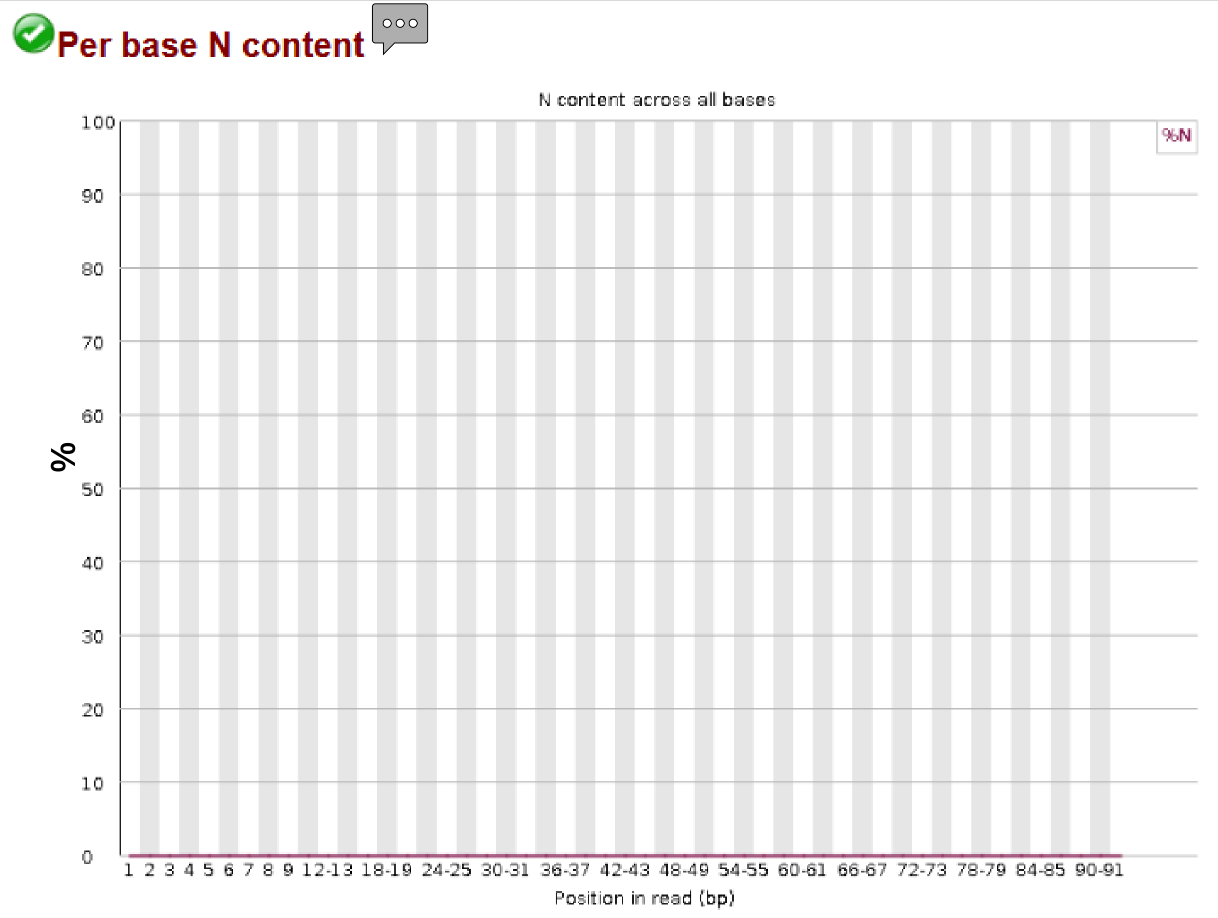
## Per sequence GC content



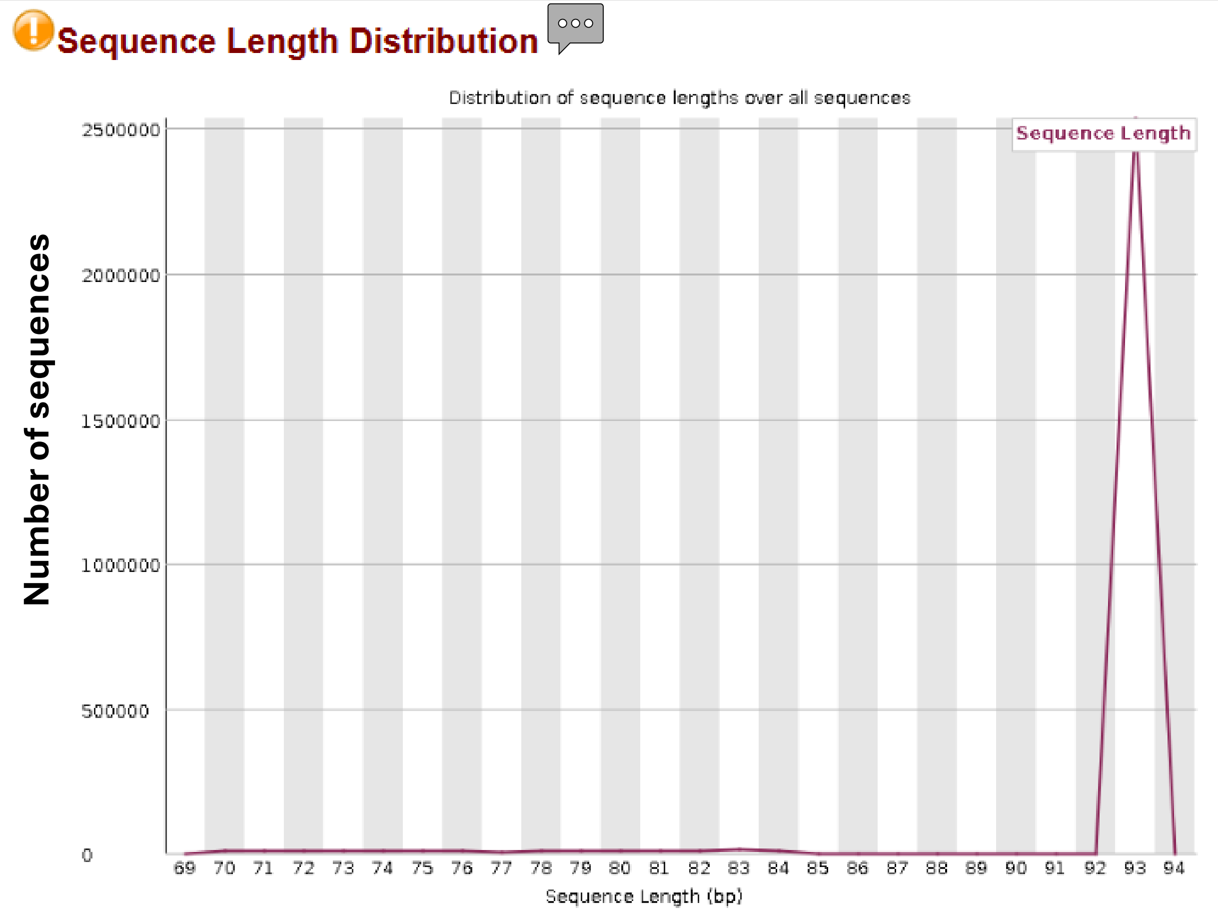

The GC content across the whole length of each sequence in a file and compares it to a modelled normal distribution of GC content. In a **normal random library,** you would expect to see a roughly normal distribution of GC content where the central peak corresponds to the overall GC content of the underlying genome.

*Answer*

# Quality Check (QC) with FastQC




This module plots out the percentage of base calls at each position for which an N was called.

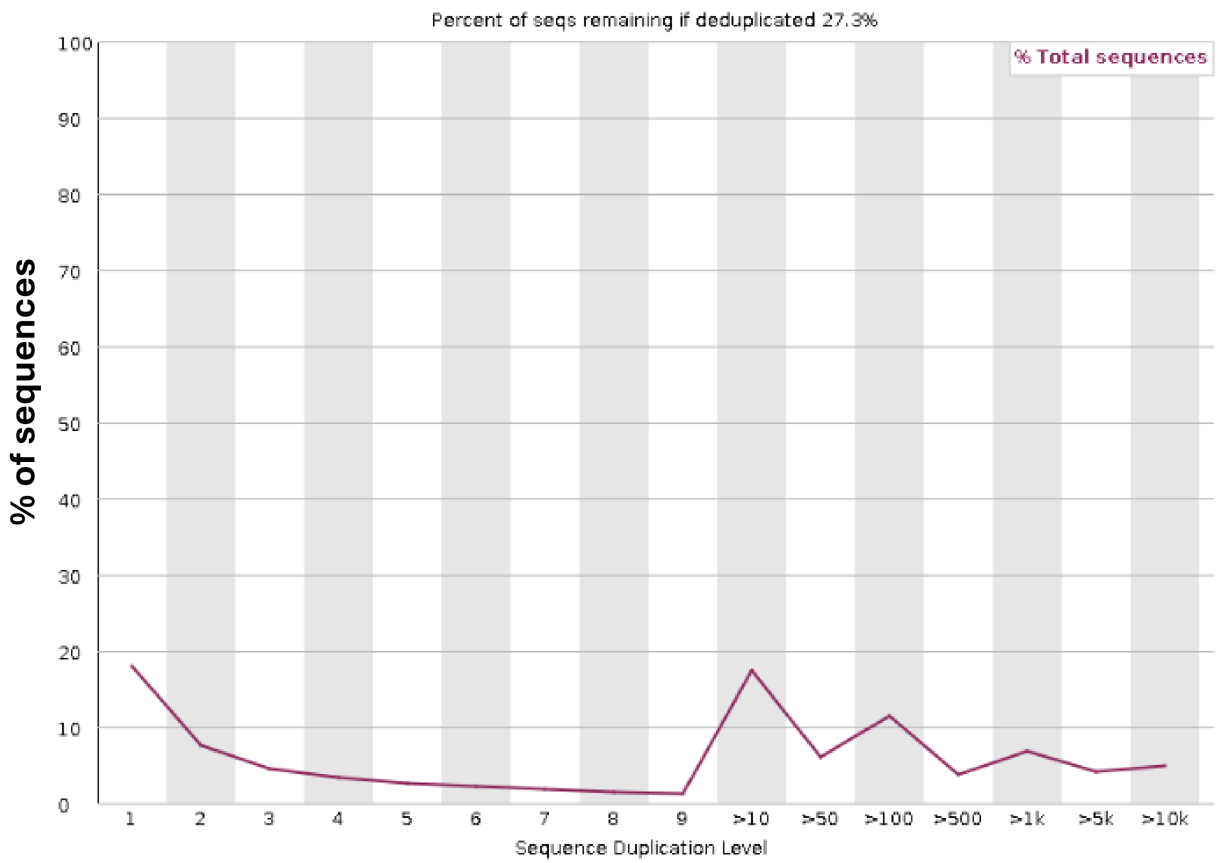



This module generates a graph showing the distribution of fragment sizes in the file which was analysed.

**Answer**

# Quality Check (QC) with FastQC

## Sequence Duplication Levels




This module counts the degree of duplication for every sequence in a library and creates a plot showing the relative number of sequences with different degrees of duplication. A high level of duplication is more likely to indicate some kind of enrichment bias. **IMPORTANT**: In RNA-Seq libraries, sequences from different transcripts will be present at wildly different levels in the starting population. In order to be able to observe lowly expressed transcripts it is therefore common to greatly over-sequence high expressed transcripts, and this will potentially create large set of duplicates.

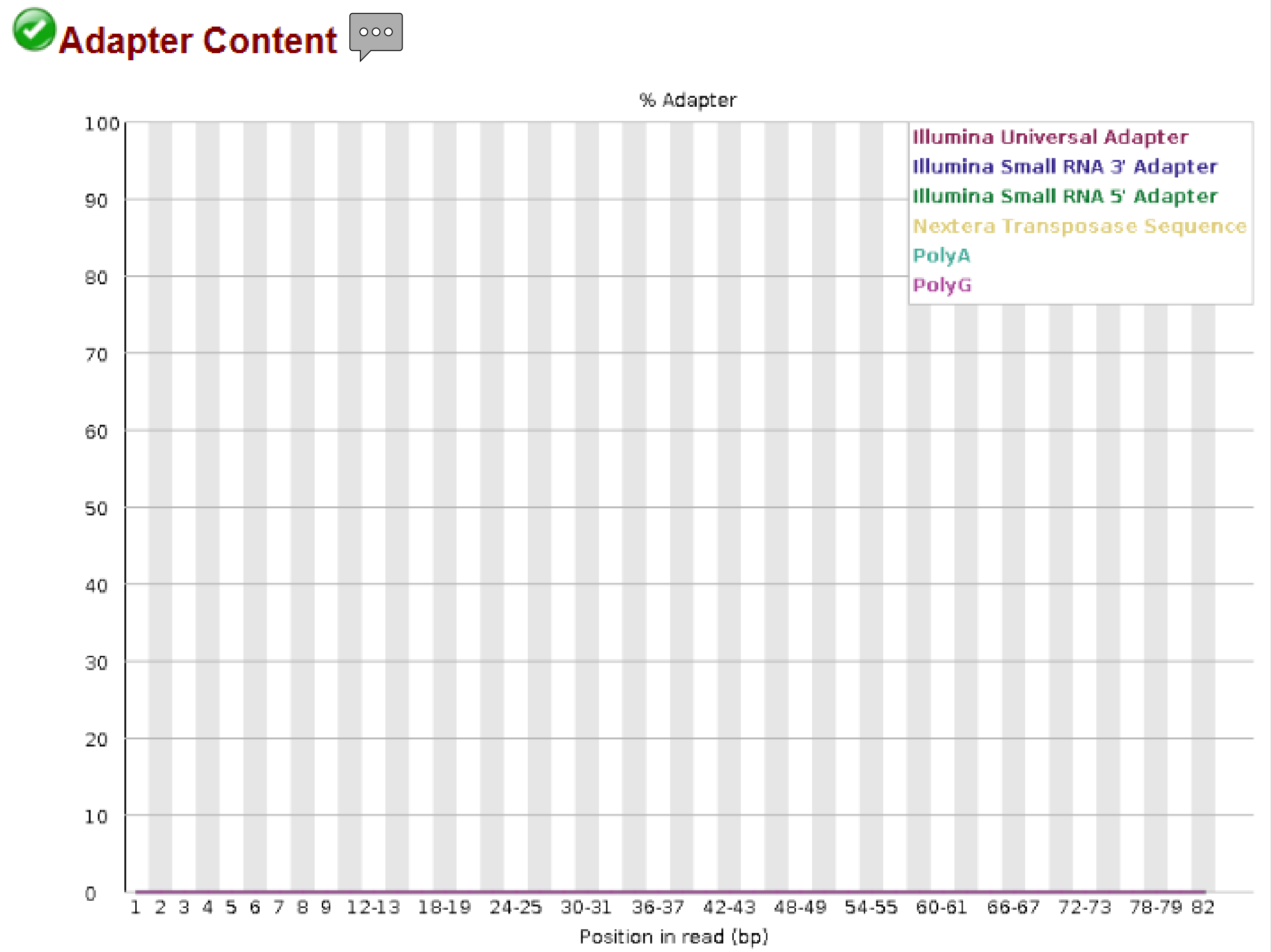
## Overrepresented sequences

| Sequence | Count | Percentage | Possible Source |
|---|---|---|---|
| GGGGAAATGCACCTGGTGCAGCAGCTAATCGAGTGGCTTTAGAAGCCTGT | 40784 | 1.5074427769414185 | No Hit |
| GGGAAATGCACCTGGTGCAGCAGCTAATCGAGTGGCTTTAGAAGCCTGTG | 19675 | 0.7272199057552571 | No Hit |
| GGGATGATTCTGTATTACAATTTGGTGGAGGAACTTTAGGACATCCTTGG | 18927 | 0.6995726127689836 | No Hit |
| GGACATCCTTGGGGAAATGCACCTGGTGCAGCAGCTAATCGAGTGGCTTT | 16776 | 0.6200681646226275 | No Hit |
| GGGCGCGATCTTGCTCGTGAAGGTAATGAAATTATCCGAGCAGCTTGCAA | 12819 | 0.47381102779550904 | No Hit |
| TACCTAGGCACCCAGAGACGAGGAAGGGCGTAGCAAGCGACGAAATGCTT | 12385 | 0.4577696840040081 | No Hit |
| GAAGGTAATGAAATTATCCGAGCAGCTTGCAAATGGAGTCCTGAACTAGC | 10784 | 0.3985941277593236 | No Hit |
| GGAGGAACTTTAGGACATCCTTGGGGAAATGCACCTGGTGCAGCAGCTAA | 9593 | 0.3545728363867945 | No Hit |
| GGTGCAGCAGCTAATCGAGTGGCTTTAGAAGCCTGTGTACAAGCTCGTAA | 9539 | 0.3525769088182667 | No Hit |
| GGGGATGATTCTGTATTACAATTTGGTGGAGGAACTTTAGGACATCCTTG | 8802 | 0.32533619367002664 | No Hit |
| GGATGATTCTGTATTACAATTTGGTGGAGGAACTTTAGGACATCCTTGGG | 8521 | 0.3149499779893543 | No Hit |
| GGGAGAGCAATACAAGCGTTGTGCTGCTAGGCGAAGCGGTTGAGTGCCGC | 8453 | 0.3124365877178749 | No Hit |
| GGTAATGAAATTATCCGAGCAGCTTGCAAATGGAGTCCTGAACTAGCCGC | 8362 | 0.30907308014868917 | No Hit |
| GGAGTATCCTTGATATATAAATATATAGATAAATAGATTTAAATGGAAAA | 8070 | 0.2982802866299835 | No Hit |
| GTATGGAAGGCGATCAAATTCGAGTTCGCGCCGGTAGATACTATCGATTA | 7873 | 0.29099884716702107 | No Hit |
| GTATTTAGCCTTGCAAGGTGGTCCTTGCTGATTCACACGGGATTCCACGT | 6178 | 0.2283488984882327 | No Hit |
| GGAGTCCTGAACTAGCCGCAGCTTGTGAAGTATGGAAGGCGATCAAATTC | 5746 | 0.21238147794001055 | No Hit |

This module lists all of the sequence which make up more than 0.1% of the total. For each overrepresented sequence the program will look for matches in a database of common contaminants and will report the best hit it finds. This module will often be triggered when used to analyse small RNA libraries where sequences are not subjected to random fragmentation, and the same sequence may naturally be present in a significant proportion of the library.

*Answer*

# Quality Check (QC) with FastQC




It checks if your library contains a significant amount of adapters in order to be able to assess whether you need to adapter trim or not. The plot itself shows a cumulative percentage count of the proportion of your library which has seen each of the adapter sequences at each position

# Filtering with `Trimmomatic`

## Task 02: Filter the sequences of ABR3_W_03 sample. Select the most appropriate parameters based on the QC report to improve their quality.

**Software:** Trimmomatic v.0.39

**Command:**
```
java -jar trimmomatic-0.39.jar SE -threads 4 -phred33 \
Input_file Output_file\
ILLUMINACLIP:path/adapter_file:x:xx:xx \
LEADING:xx TRAILING:xx \
SLIDINGWINDOW:xx:xx \
MINLEN:xx
```

## Tips:

- ✓ Check that the paths of the software, input files, and output folder are correct.
- ✓ *Phred+33 is listed as Illumina 1.9/Sanger (see QC report)*
- ✓ *Adapter file TruSeq3-SE.fa for Illumina HiSeq2500*
- ✓ Check Trimmomatic website: http://www.usadellab.org/cms/?page=trimmomatic

*Answer*

## Filtering with Trimmomatic

```
java -jar trimmomatic-0.39.jar \
SE \
-threads 4 \
-phred33 \
01_RNAseq_raw/ABR3_W_03.fastq.gz \
02_RNAseq_filtered/ABR3_W_03_trimmed.fastq.gz \
ILLUMINACLIP:software/Trimmomatic-0.39/adapters/TruSeq3-SE.fa:2:30:10 \
LEADING:3 \
TRAILING:3 \
SLIDINGWINDOW:4:15 \
MINLEN:36
```

## Quality Check (QC) after filtering step

**Task 03: Generate a new QC report of the filtered sequences and compare the reports before and after filtering. What parameters have changed?**

**Software:** FastQC v.0.12.1

***Answer***

# Comparing QC reports before and after filtering step

```
fastqc 02_RNAseq_filtered/ABR3_W_03_trimmed.fastq.gz  -o 02_RNAseq_filtered/reports/
```

***Before ...***

## Basic Statistics

| Measure | Value |
|---|---|
| Filename | ABR3_W_03.fastq.gz |
| File type | Conventional base calls |
| Encoding | Sanger / Illumina 1.9 |
| Total Sequences | 2705509 |
| Total Bases | 248.8 Mbp |
| Sequences flagged as poor quality | 0 |
| Sequence length | 70-93 |
| %GC | 45 |

***After ...***

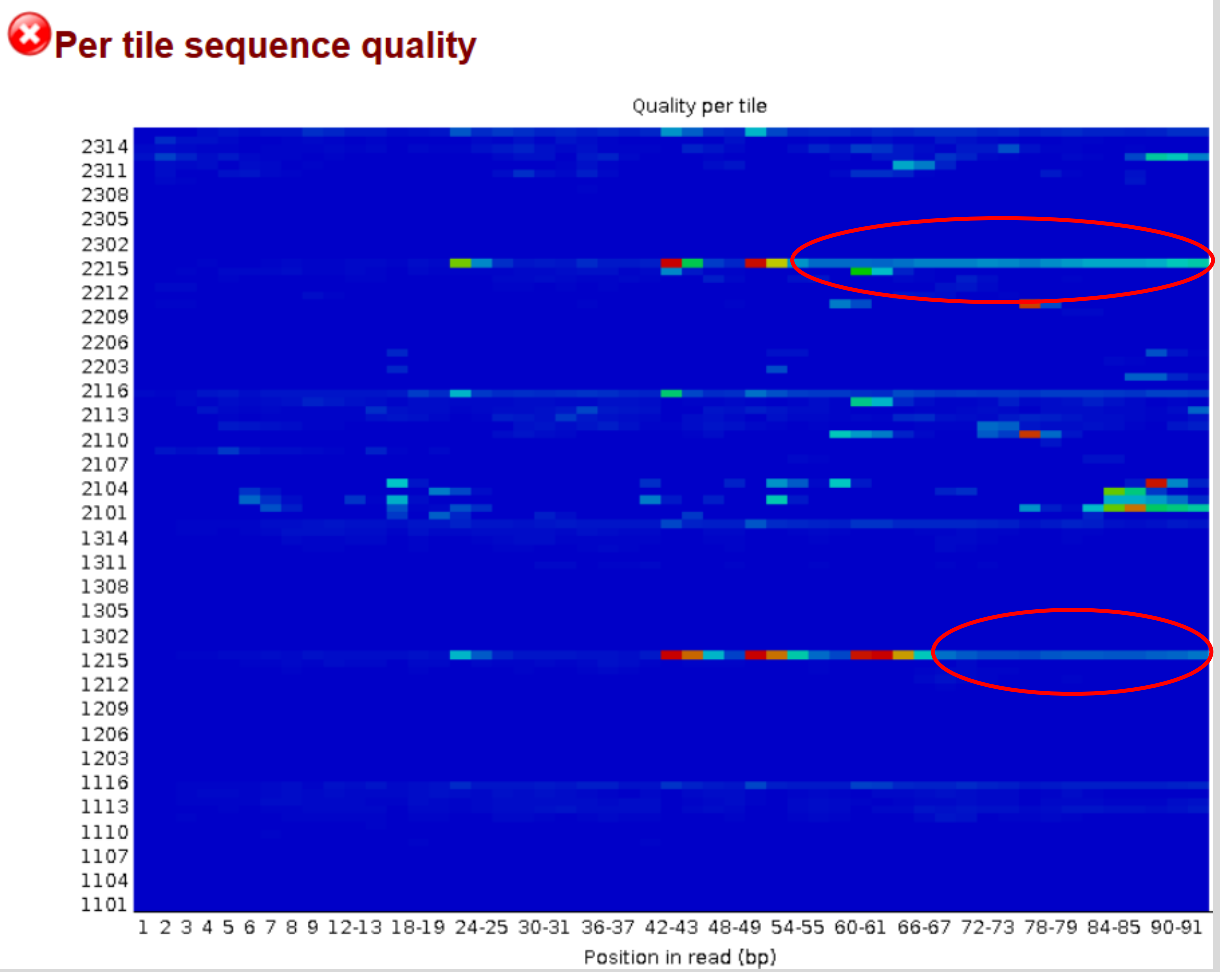
## Basic Statistics

| Measure | Value |
|---|---|
| Filename | ABR3_W_03_trimmed.fastq.gz |
| File type | Conventional base calls |
| Encoding | Sanger / Illumina 1.9 |
| Total Sequences | 2666619 |
| Total Bases | 241.2 Mbp |
| Sequences flagged as poor quality | 0 |
| Sequence length | 36-93 |
| %GC | 45 |

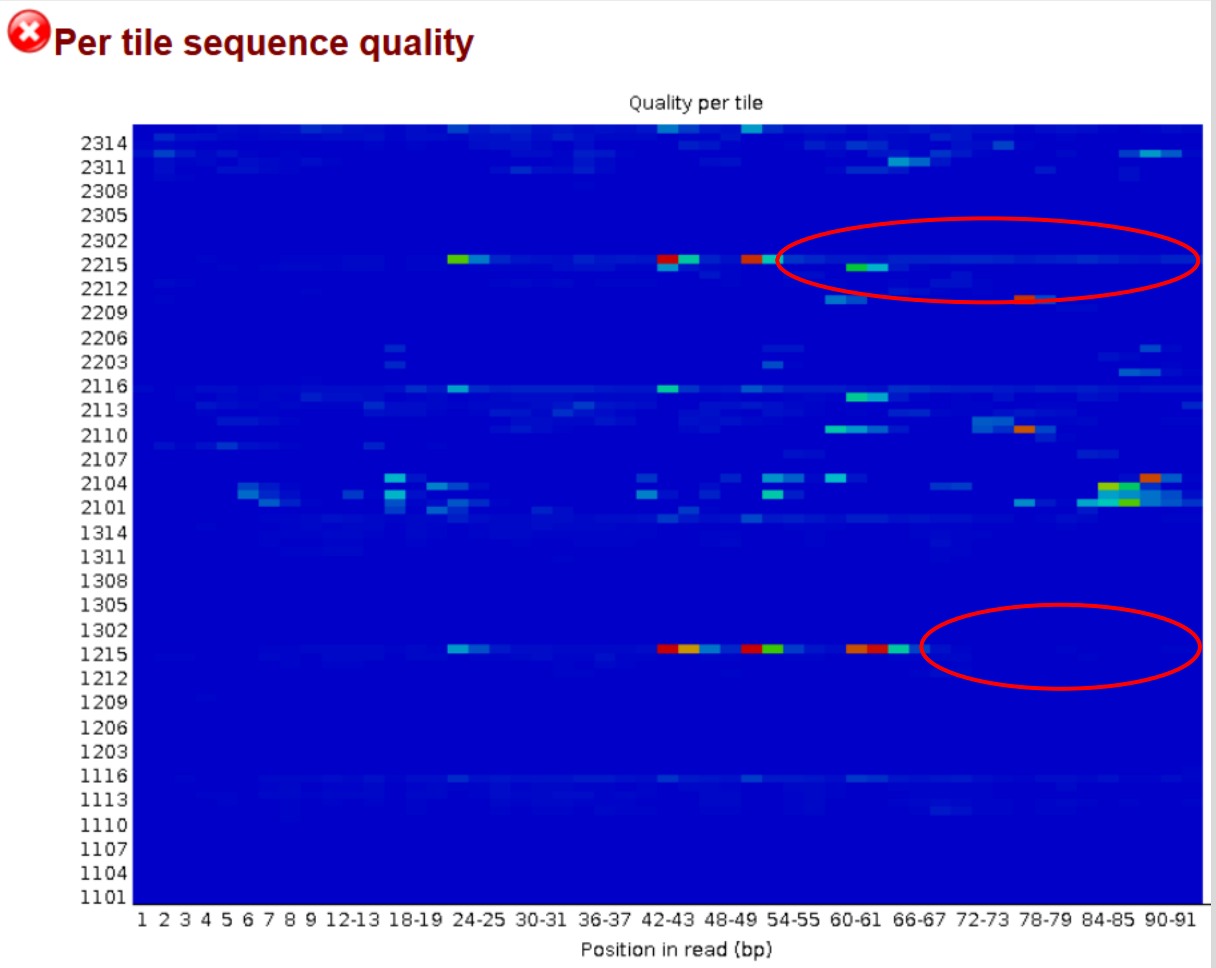
**Answer**

# Comparing QC reports before and after filtering step

***Before …***




***After …***



# Mapping with kallisto

**Task 04: Perform pseudo-alignments of the reads from sample ABR3_W_03 against the reference transcriptome (`Bdistachyon_314_v3.1.transcript.TagSeq500b.fa.gz`) and indicate the expression of the Bradi3g51200 gene in est_counts and TPM.**

**Software:** kallisto 0.48

```
# Index reference
kallisto index \
-i ref.output.idx ref.input.fa.gz

# Quantification
kallisto quant \
-i ref.output.idx \
-o output_folder/output_subfolder_sample \
-b 100 \            # Number of bootstrap samples*
--single \          # Quantify single-end reads
-l 100 \            # Estimated average fragment length**
-s 20 \             # Estimated standard deviation of fragment length***
<(zcat input_compressed.fastq.gz)
```

# Mapping with kallisto

```
* Number of bootstrap samples
** Estimated average fragment length
*** Estimated standard deviation of fragment length
```

**Number of bootstrap samples (-b):** Because many reads can map to multiple transcripts, kallisto uses a probabilistic model to assign them. To measure how uncertain these assignments are, kallisto performs bootstrap resampling — it repeatedly resamples reads (with replacement) and re-estimates transcript abundances. Typical bootstrapping value: 100

**Estimated average fragment length (-l):** In the case of single-end reads, the -l option must be used to specify the average fragment length. Typical Illumina libraries produce fragment lengths ranging from 180–200 bp but it's best to determine this from a library quantification with an instrument such as an Agilent Bioanalyzer.
For paired-end reads, the average fragment length can be directly estimated from the reads, and the program will do so if -l is not used (this is the preferred run mode).

**Estimated standard deviation of fragment length (-s):** This value describes how much variation there is in the length of the sequenced cDNA fragments. Library kits or sequencing facilities often report the average fragment size and its variation, e.g. "200 ± 20 bp" → -l 200 -s 20. Typical 's' value: 20

***Answer***

# Mapping with kallisto

**Pseudoalignments** of RNA-Seq data were carried out using as references 500 bp from the 3' tails of the *B. distachyon*_314_version 3.1 **transcriptome** (IBI, 2010; https://phytozome.jgi.doe.gov/).

We applied estimated average fragment lengths of 100 bp and standard deviations of fragment length of 20.

```
# Index reference
kallisto index \
-i 00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.idx \
00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.fa.gz

# Quantification
kallisto quant \
-i 00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.idx \
-o 03_quant/ABR3_W_03 \
-b 100 \            # Number of bootstrap samples
--single \          # Quantify single-end reads
-l 100 \            # Estimated average fragment length
-s 20 \             # Estimated standard deviation of fragment length
<(zcat 02_RNAseq_filtered/ABR3_W_03_trimmed.fastq.gz)
```

# Mapping with kallisto

**Output files:**

- **`abundance.h5`:** a HDF5 binary file containing run info, abundance estimates, bootstrap estimates, and transcript length information length. This file can be read in by `sleuth`.
- **`abundance.tsv`:** a plaintext file of the abundance estimates. It does not contain bootstrap estimates. The first line contains a header for each column, including estimated counts, TPM, effective length.
- **`run_info.json`:** a json file containing information about the run.

**abundance.tsv**

| target_id | length | eff_length | est_counts | tpm |
|---|---|---|---|---|
| Bradi0012s00100.1 | 500 | 401 | 34.4331 | 18.6309 |
| Bradi0012s00201.1 | 500 | 401 | 0.000217899 | 0.0001179 |
| Bradi0012s00201.2 | 500 | 401 | 0.000217899 | 0.0001179 |
| Bradi0012s00201.3 | 500 | 401 | 0.000217899 | 0.0001179 |
| Bradi0012s00201.4 | 500 | 401 | 2.2899 | 1.23901 |
| Bradi0012s00201.5 | 500 | 401 | 0 | 0 |
| Bradi0014s00100.1 | 500 | 401 | 0 | 0 |
| Bradi0135s00100.1 | 500 | 401 | 0 | 0 |
| Bradi0180s00100.1 | 500 | 401 | 0 | 0 |
| Bradi1g00200.1 | 500 | 401 | 0 | 0 |
| Bradi1g00210.3 | 500 | 401 | 0 | 0 |
| Bradi1g00213.1 | 500 | 401 | 0 | 0 |
| Bradi1g00215.1 | 500 | 401 | 0 | 0 |
| Bradi1g00217.1 | 381 | 282 | 0 | 0 |
| Bradi1g00219.1 | 500 | 401 | 1 | 0.541076 |
| Bradi1g00222.1 | 324 | 225 | 13.952 | 13.4541 |

**run_info.json**


```
{
        "n_targets": 52972,
        "n_bootstraps": 100,
        "n_processed": 2666619,
        "n_pseudoaligned": 1828214,
        "n_unique": 1262059,
        "p_pseudoaligned": 68.6,
        "p_unique": 47.3,
        "kallisto_version": "0.48.0",
        "index_version": 10,
        "start_time": "Thu Oct 30 15:01:55 2025",
        "call": "kallisto quant -i 00_Reference_transcriptome/Bdistachyon_314_v3.1.transcript.TagSeq500b.idx -o 03_quant/ABR3_W_03 -b 100 --single -l 100 -s 20
 /dev/fd/63"
}
```

*Answer*

# Mapping with kallisto

**Gene expression fo Bradi3g51200 (sample ABR3_W_03):**

✓ `est_counts = 60`

✓ `TPM = 32.4646`

## Readme (Sleuth and WGCNA)

- ✓ Into `/data` (Ubuntu WSL)
  - `cp -r /home/vep/test_data/transcripts/brachy/05_WGCNA/ .`
  - `cp -r /home/vep/test_data/transcripts/brachy/04_Sleuth/ .`
  - Extent permissions: `chmod -R 777 04_Sleuth 05_WGCNA`
  - Copy Rstudio scripts:
    - Copy Rstudio script "`Practical_Sleuth.rmd`" in

      \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\04_Sleuth
    - Copy Rstudio script "`Practical_WGCNA_W_dataset.rmd`" in

      \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\05_WGCNA
    - Copy Rstudio script "`Practical_WGCNA_D_dataset.rmd`" in

      \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\05_WGCNA

# Data exploration and DEGs with Sleuth

**Task 05: Run the Sleuth pipeline to explore your data and DEGs.**

**Software:** `sleuth v.0.30.2`

**Pipeline in Rstudio:** `Practical_Sleuth.Rmd`

In \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\04_Sleuth

**Task 05.1: Use sleuth_live(so) to explore the boxplots of the Bradi1g06560.1 isoform/transcript expression (in TPMs) with respect to the experimental conditions (drought and irrigation).**

# Answer 05.1

## Data exploration and DEGs with Sleuth

***Sleuth_live(so): analyses → transcript view***

sleuth overview analyses maps summaries diagnostics settings

### transcript view

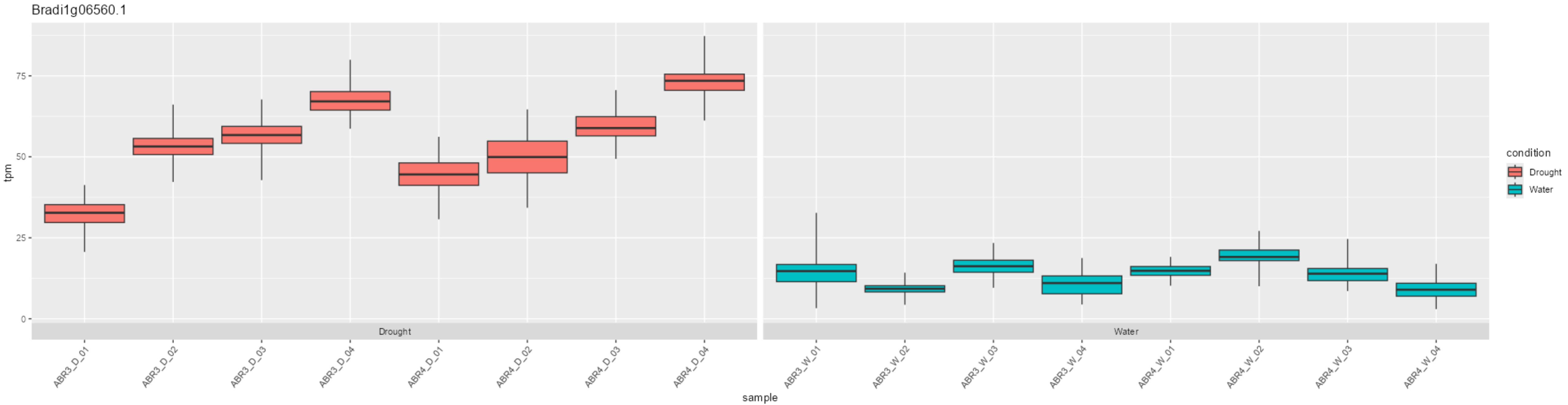
Boxplots of transcript abundances showing technical variation in each sample.

**transcript:** ? Bradi1g06560.1

**color by:** ? condition

**units:** ? tpm

view



Data exploration and DEGs with `Sleuth`

**Task 05.2: Use sleuth_live(so) to explore the heatmap of the following isoforms: Bradi1g06560.1 Bradi2g54810.2 Bradi4g08240.1 Bradi4g31150.1. Use as covariate “condition” and as units TPMs.**

# Answer 05.2

# Data exploration and DEGs with Sleuth

***Sleuth_live(so): analyses → heatmap***

sleuth overview analyses ▾ maps ▾ summaries ▾ diagnostics ▾ settings

## heat map

Plot of select abundances in a clustered heat map using the R hclust function. The heat map iteratively groups together transcripts with similar expression patterns, highlighting similarities among the transcripts themselves. For a more in-depth discussion of hierarchical clustering, see 10.1007/BF02289588.

Enter space-separated values (The ten transcripts with the lowest q-values are already entered. Click view to see them!).

**enter target ids:** ?

Bradi1g06560.1 Bradi2g54810.2 Bradi4g082

**covariates:** ?

☐ genotype ☑ condition ☐ rep ☐ combinated

**units:** ?

tpm

**transform:** ?

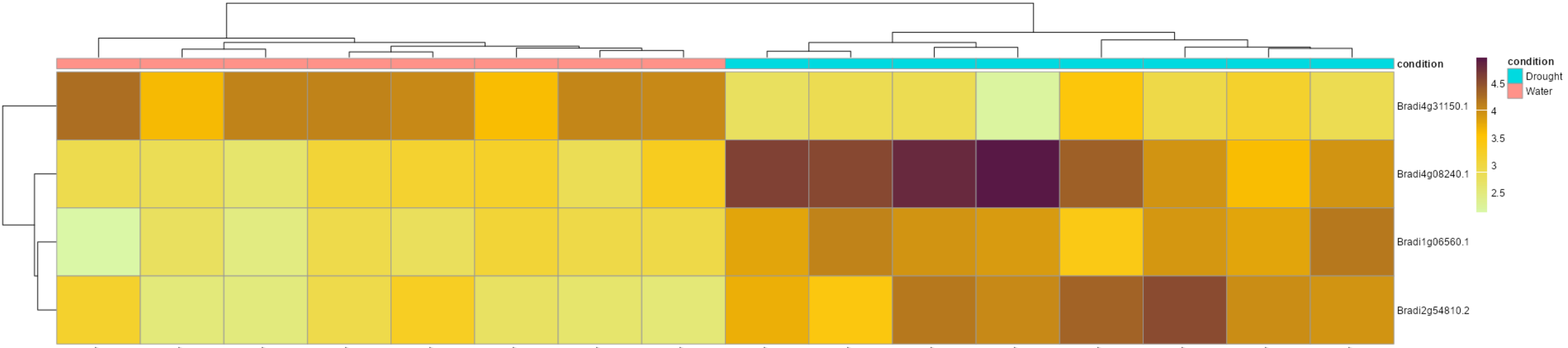
log

**offset:** ?

1

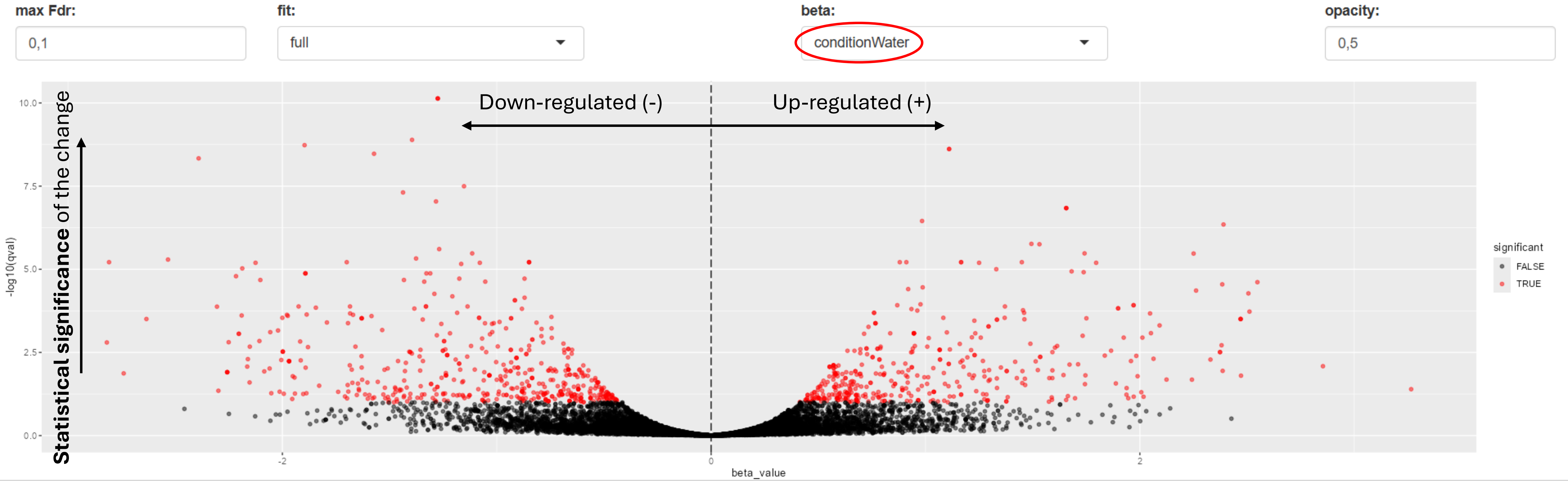
☑ cluster transcripts ?

view



# Data exploration and DEGs with Sleuth

***Sleuth_live(so): analyses → volcano plot***




**X-axis (log2 fold change)** This represents the **magnitude of change in gene expression** between two conditions.

# Data exploration and DEGs with Sleuth

## *Sleuth_live(so): summaries → densities*

sleuth overview analyses maps summaries diagnostics settings

## distribution of abundances

Distributions of abundances of individual samples or groupings by covariates.

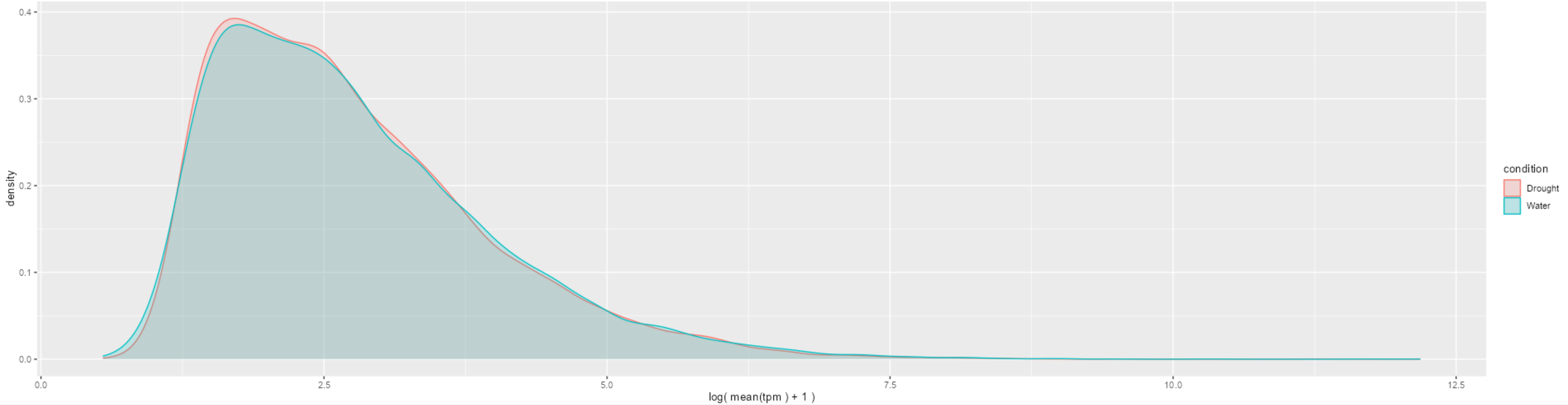
**grouping:** condition

**units:** ? tpm

filter ?

**transform:** ? log

**offset:** ? 1

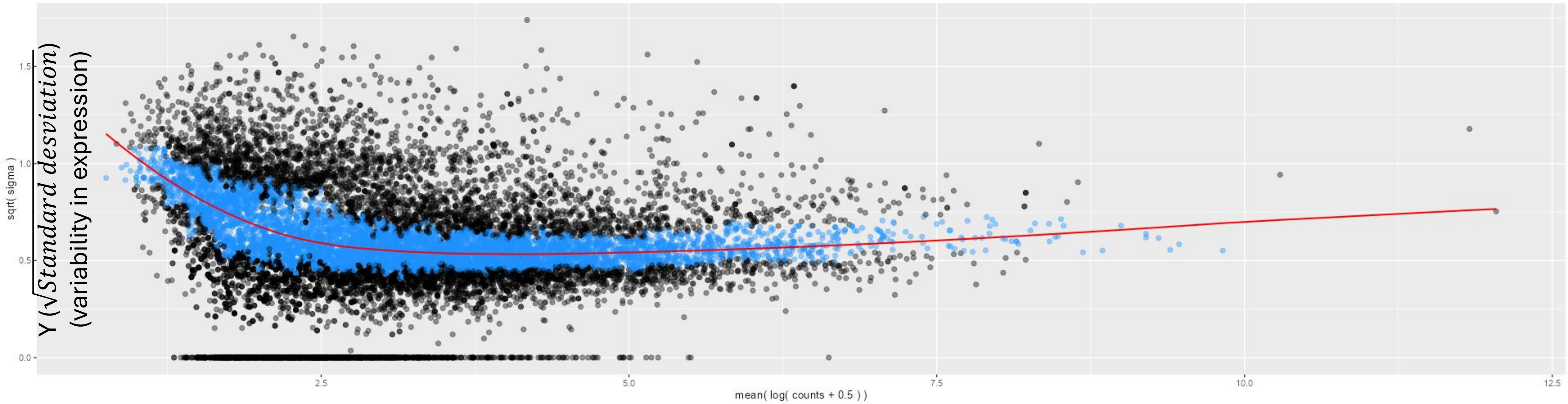


# Data exploration and DEGs with Sleuth

***Sleuth_live(so): diagnostics → mean-variance plot***

## mean-variance plot

Plot of abundance versus square root of standard deviation which is used for shrinkage estimation. The blue dots are in the interquartile range and the red curve is the fit used by sleuth. Any points at y = 0 have inferential variance greater than observed total raw variance.




X (Abundance)

# Data exploration and DEGs with Sleuth

## *Sleuth_live(so): diagnostics → Q-Q plot*

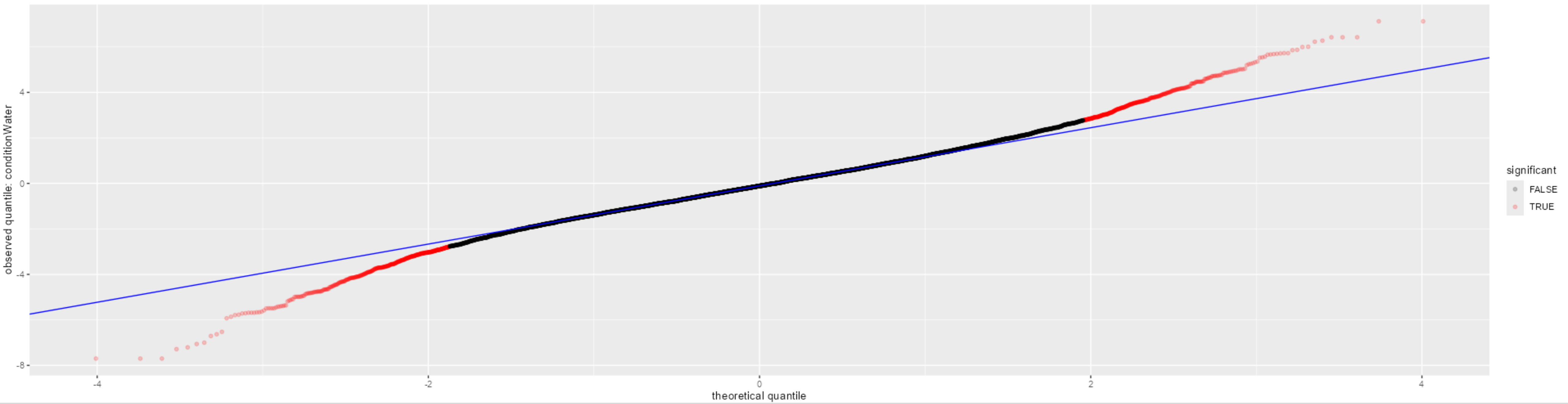
## Q-Q plot

Select the test and view the appropriate quantile-quantile plot. Points that are color red are considered "significant" at the selected fdr-level.

**max Fdr:** 0,1

**fit:** full

**beta:** conditionWater



## Data exploration and DEGs with `Sleuth`

- ✓ **Sleuth** is a powerful R package designed to analyze gene expression values calculated by the **Kallisto** software (pseudo-aligner).
- ✓ **Sleuth** allows for **exploring** and **visualizing** the data, which is essential for this type of analysis.
- ✓ **Sleuth** also allows for the identification of **differentially expressed genes (DEGs)**.

# Co-expression Network with WGCNA

**Task 05: Run the WGCNA pipeline to construct and explore the co-expression network.**

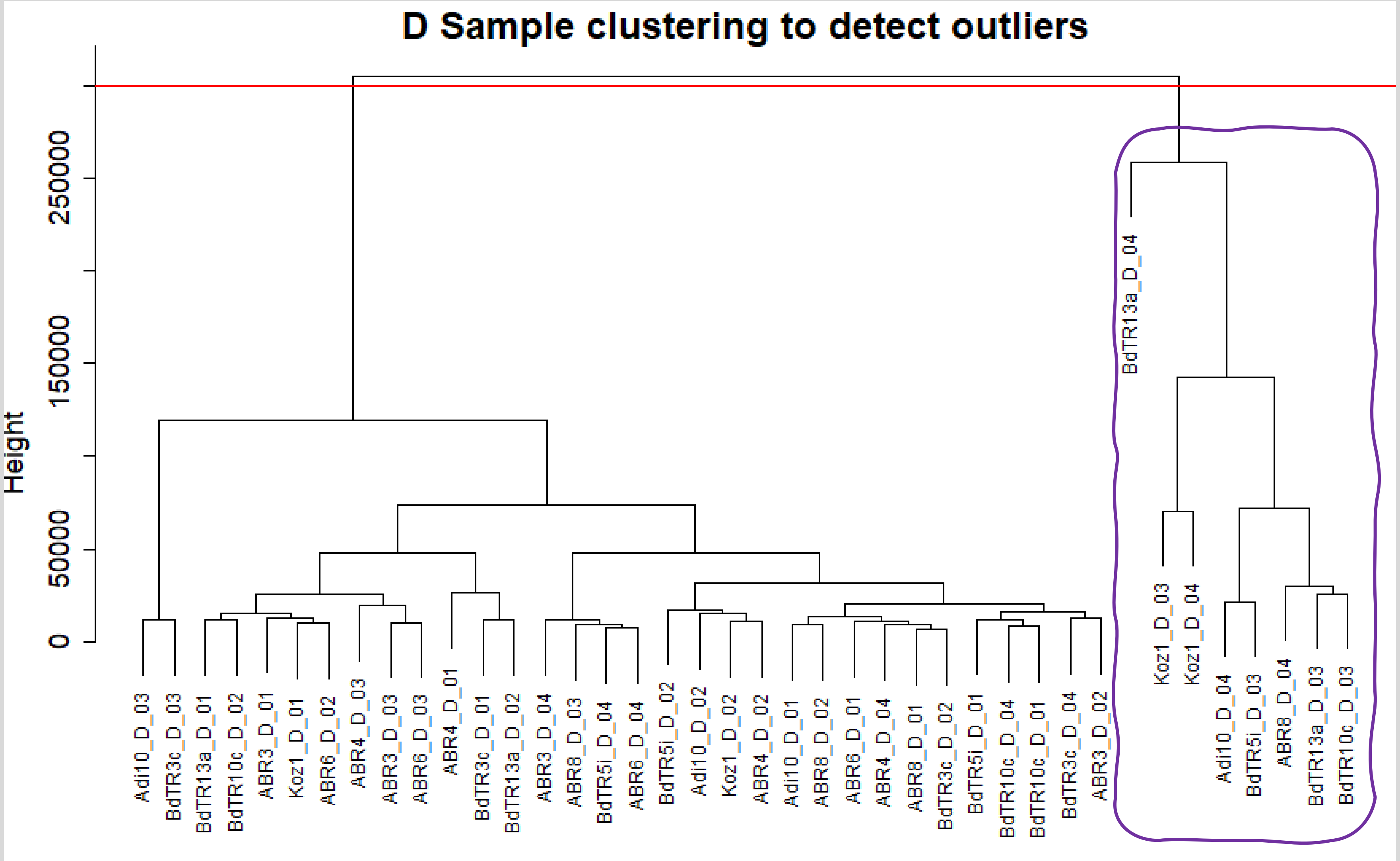
**Software:** `WGCNA v.1.73`

**Pipeline in Rstudio:** `Practical_WGCNA.Rmd`

In \\wsl.localhost\Ubuntu-22.04\home\iamz\transcriptome_class\05_WGCNA

# Co-expression Network with WGCNA

## Outliers Detection

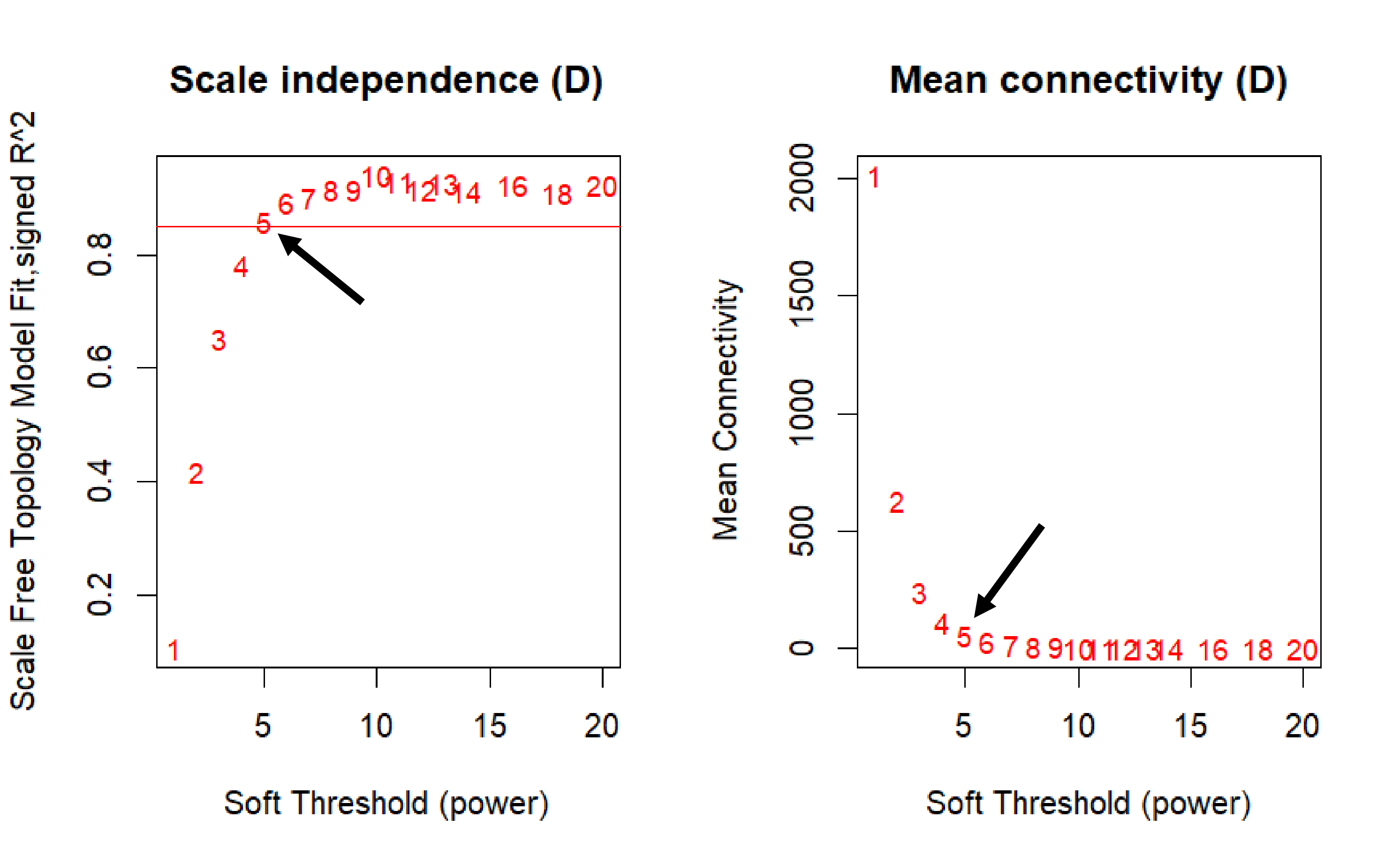


# Co-expression Network with WGCNA

Soft-thresholding powers selection for network construction

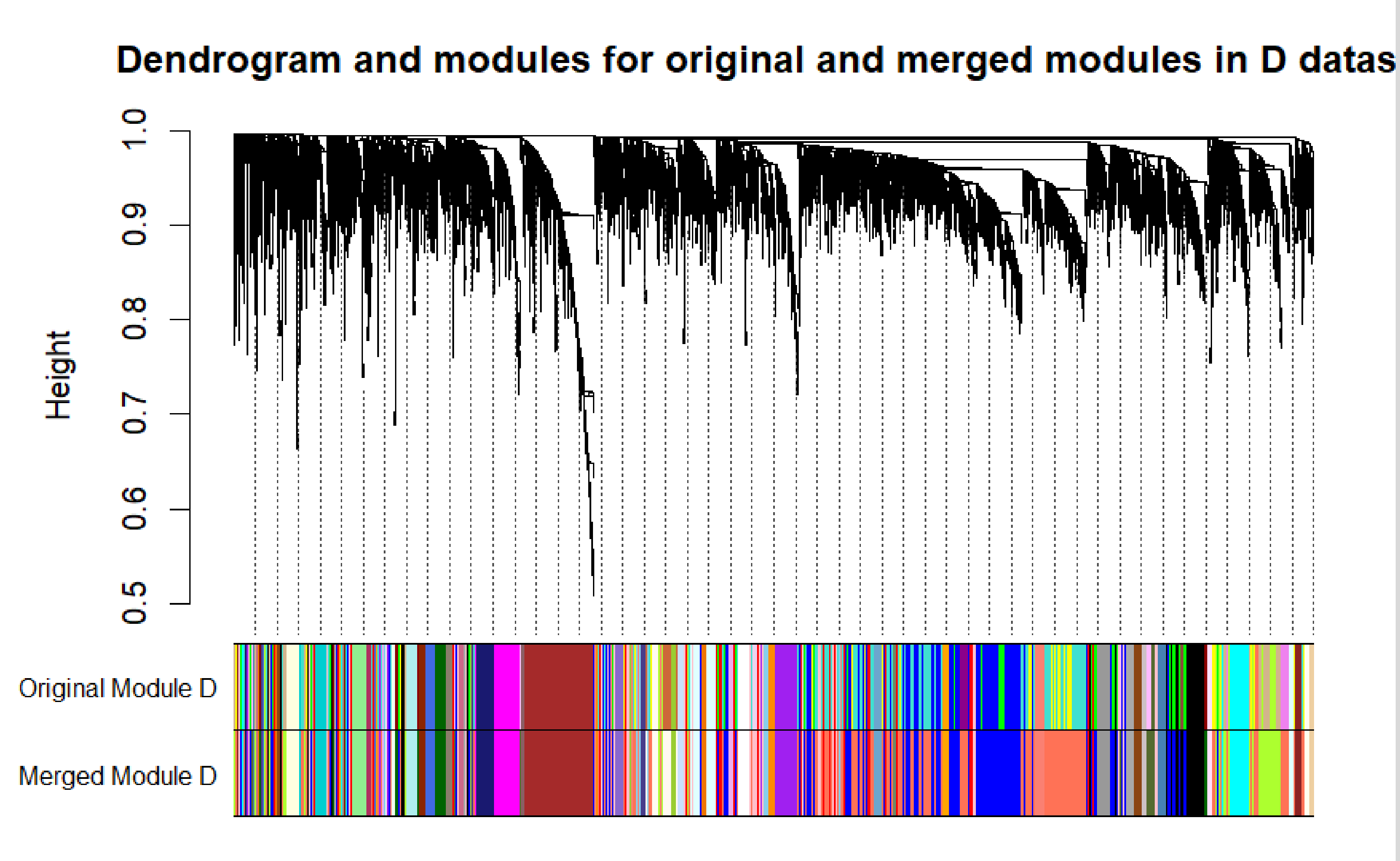
$$a_{ij} = s_{ij}^{\beta},$$

| Power <dbl> | SFT.R.sq <dbl> | mean.k. <dbl> |
|---|---|---|
| 1 | 0.106 | 2010.000 |
| 2 | 0.417 | 627.000 |
| 3 | 0.652 | 243.000 |
| 4 | 0.782 | 109.000 |
| 5 | 0.858 | 54.800 |
| 6 | 0.891 | 29.900 |
| 7 | 0.902 | 17.500 |
| 8 | 0.914 | 10.900 |
| 9 | 0.916 | 7.230 |
| 10 | 0.941 | 5.030 |



# Co-expression Network with WGCNA

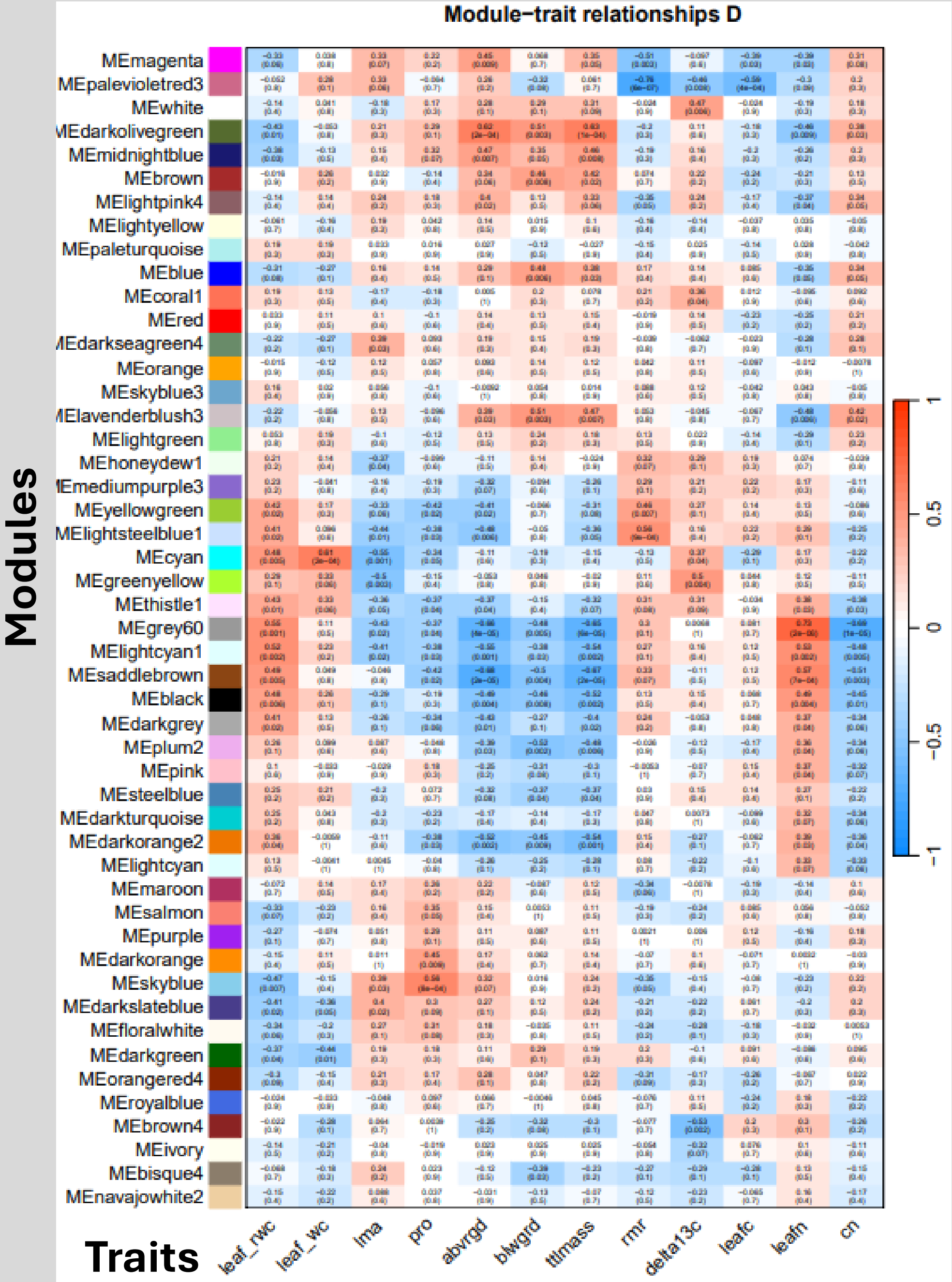
## Network construction and Module detection




The ***gray*** module represents unclustered genes (without an assigned module).
**Note**: In this exercise, very few genes appear in "gray" because I filtered them beforehand to reduce the database size and speed up computation.

# Co-expression Network with WGCNA

## Module-trait associations



# Co-expression Network with WGCNA

**Task 06. Perform a Statistical overrepresentation test (https://pantherdb.org/) to determine if the genes of the magenta module of the drought network are significantly involved in any biological process according to the "GO“**

**(The format of the ID gene must be converted: sed 's/Bradi/BRADI_/g’ ID.txt | sed -E 's/\.[0-9]/v3/g' | sort | uniq | tr '\n' ' ')**

BRADI_1g01507v3 BRADI_1g03590v3 BRADI_1g03610v3 BRADI_1g04440v3 BRADI_1g07237v3 BRADI_1g08410v3 BRADI_1g08870v3 BRADI_1g09950v3 BRADI_1g11630v3
BRADI_1g11960v3 BRADI_1g13232v3 BRADI_1g14940v3 BRADI_1g15030v3 BRADI_1g16190v3 BRADI_1g16620v3 BRADI_1g20360v3 BRADI_1g20620v3 BRADI_1g20930v3
BRADI_1g26260v3 BRADI_1g27490v3 BRADI_1g29040v3 BRADI_1g30247v3 BRADI_1g32330v3 BRADI_1g32660v3 BRADI_1g32990v3 BRADI_1g33300v3 BRADI_1g35742v3
BRADI_1g36290v3 BRADI_1g37080v3 BRADI_1g37175v3 BRADI_1g37750v3 BRADI_1g38760v3 BRADI_1g42730v3 BRADI_1g44047v3 BRADI_1g44230v3 BRADI_1g45080v3
BRADI_1g45160v3 BRADI_1g45200v3 BRADI_1g46530v3 BRADI_1g47767v3 BRADI_1g48010v3 BRADI_1g50370v3 BRADI_1g51610v3 BRADI_1g53320v3 BRADI_1g53850v3
BRADI_1g55500v3 BRADI_1g55620v3 BRADI_1g61040v3 BRADI_1g61170v3 BRADI_1g64850v3 BRADI_1g65480v3 BRADI_1g66800v3 BRADI_1g67040v3 BRADI_1g67290v3
BRADI_1g68720v3 BRADI_1g69400v3 BRADI_1g70700v3 BRADI_1g72740v3 BRADI_1g72950v3 BRADI_1g73120v3 BRADI_1g74300v3 BRADI_1g76950v3 BRADI_1g77637v3
BRADI_2g00831v3 BRADI_2g02110v3 BRADI_2g02400v3 BRADI_2g05640v3 BRADI_2g07040v3 BRADI_2g12830v3 BRADI_2g13110v3 BRADI_2g13870v3
BRADI_2g13870v30 BRADI_2g13870v33 BRADI_2g14411v3 BRADI_2g15950v3 BRADI_2g16660v3 BRADI_2g17200v3 BRADI_2g19540v3 BRADI_2g22140v3
BRADI_2g23040v3 BRADI_2g23250v3 BRADI_2g24230v3 BRADI_2g25613v3 BRADI_2g27900v3 BRADI_2g31576v3 BRADI_2g33360v3 BRADI_2g33682v3 BRADI_2g36000v3
BRADI_2g39890v3 BRADI_2g40942v3 BRADI_2g41117v3 BRADI_2g42227v3 BRADI_2g45460v3 BRADI_2g45840v3 BRADI_2g47530v3 BRADI_2g49280v3 BRADI_2g49660v3
BRADI_2g50300v3 BRADI_2g51240v3 BRADI_2g52470v3 BRADI_2g53020v3 BRADI_2g54570v3 BRADI_2g55490v3 BRADI_2g55960v3 BRADI_2g56190v3 BRADI_2g56490v3
BRADI_2g56500v3 BRADI_2g56710v3 BRADI_2g57220v3 BRADI_2g57230v3 BRADI_2g58835v3 BRADI_2g61767v3 BRADI_3g02640v3 BRADI_3g03532v3 BRADI_3g05530v3
BRADI_3g06107v3 BRADI_3g15890v3 BRADI_3g19893v3 BRADI_3g22980v3 BRADI_3g26930v3 BRADI_3g26967v3 BRADI_3g27800v3 BRADI_3g28450v3 BRADI_3g28700v3
BRADI_3g29700v3 BRADI_3g29797v3 BRADI_3g30380v3 BRADI_3g33544v3 BRADI_3g33850v3 BRADI_3g35000v3 BRADI_3g36370v3 BRADI_3g37047v3 BRADI_3g37260v3
BRADI_3g37790v3 BRADI_3g37820v3 BRADI_3g38220v3 BRADI_3g38670v3 BRADI_3g39440v3 BRADI_3g41490v3 BRADI_3g43890v3 BRADI_3g44947v3 BRADI_3g44950v3
BRADI_3g45030v3 BRADI_3g45206v3 BRADI_3g45940v3 BRADI_3g48240v3 BRADI_3g48590v3 BRADI_3g49400v3 BRADI_3g50110v3 BRADI_3g50420v3 BRADI_3g51600v3
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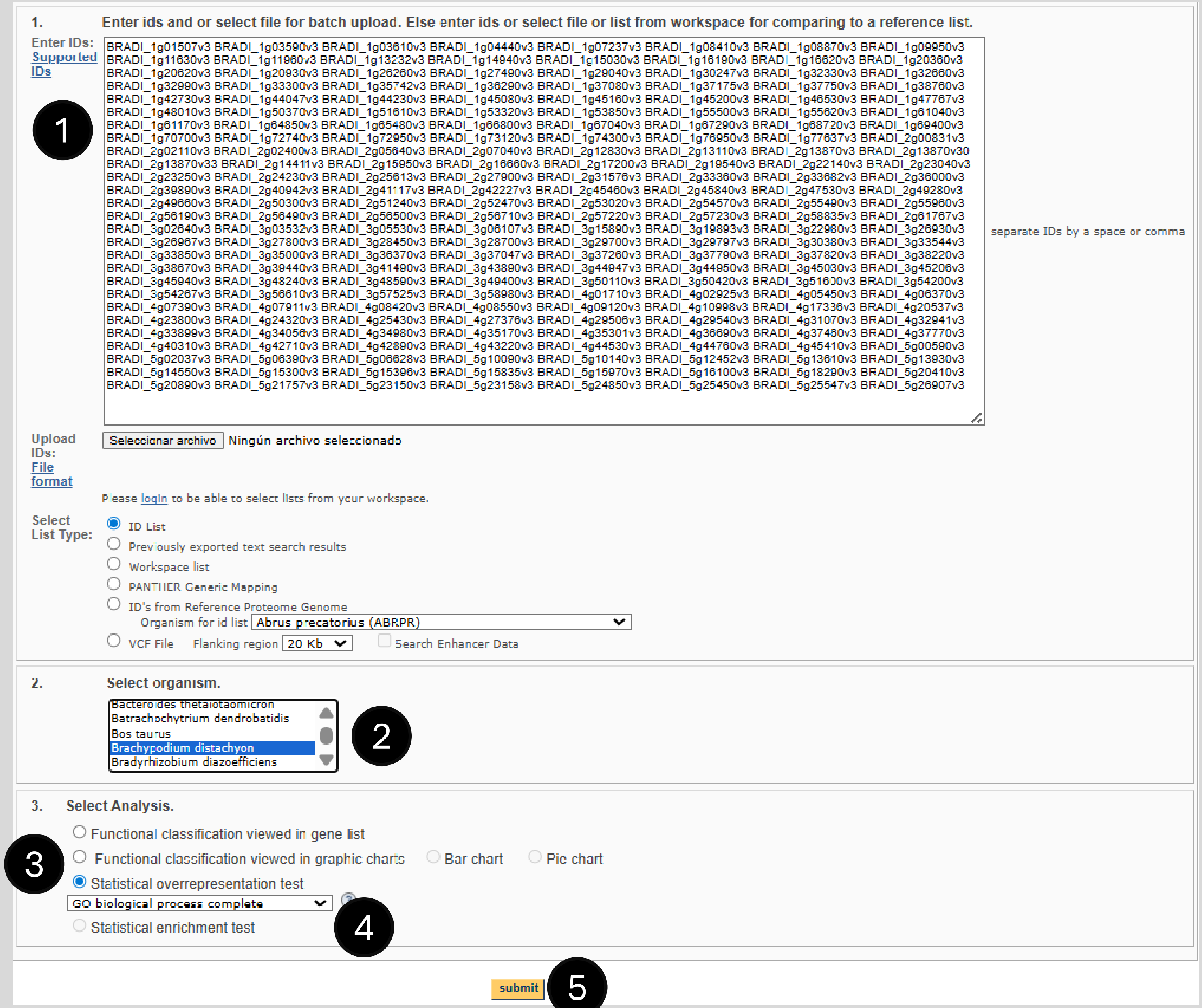
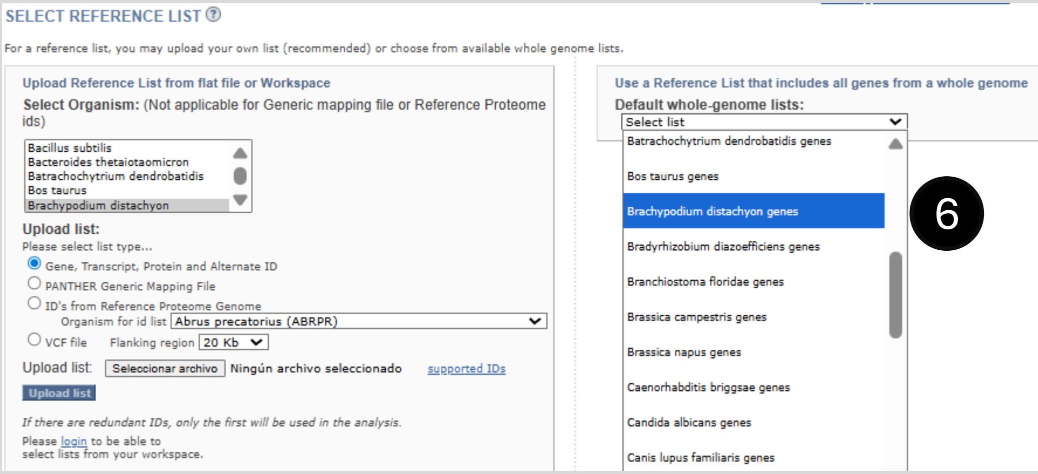
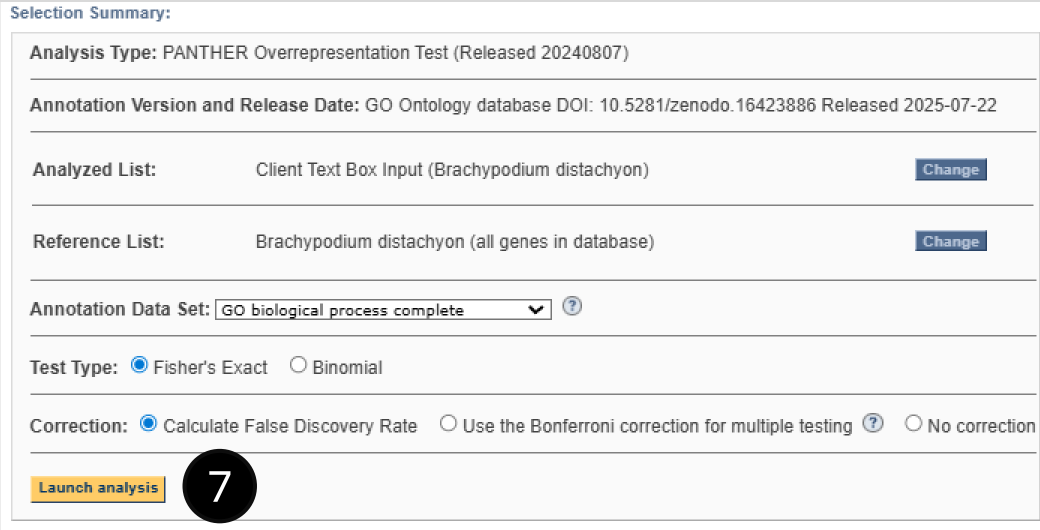
# Co-expression Network with WGCNA



# EXTRA EXCERCISES

## EXTRA EXCERCISES

**Task 07. Normalize the raw gene expression counts obtained from single-end read mappings to a reference genome, converting them into RPKM and TPM values. There are four genes (A, B, C and D) and three samples (1, 2 and 3).**

| Gene | Sample 1 | | | Sample 2 | | | Sample 3 | | |
|---|---|---|---|---|---|---|---|---|---|
| | *Raw* | *RPKM* | *TPM* | *Raw* | *RPKM* | *TPM* | *Raw* | *RPKM* | *TPM* |
| A (2,000 bp) | 15 | $0.15*10^6$ | $0.32*10^6$ | 18 | | | 35 | | |
| B (4,000 bp) | 25 | | | 30 | | | 70 | | |
| C (1,000 bp) | 10 | | | 12 | | | 20 | | |
| D (10,000 bp) | 0 | | | 1 | | | 1 | | |
| **Total reads** | **50** | | | **61** | | | **126** | | |

$$RPKM_i \text{ or } FPKM_i = \frac{q_i}{\frac{l_i}{10^3} * \frac{\sum_j q_j}{10^6}} = \frac{q_i}{l_i * \sum_j q_j} * 10^9$$

e.g. GeneA.Sample1 = $\frac{15}{\frac{2000}{10^3} * \frac{50}{10^6}} = \frac{15}{2000*50} * 10^9 = 0.15 * 10^6\ RPKM$

$$TPM_i = \frac{q_i / l_i}{\sum_j (q_j / l_j)} * 10^6$$

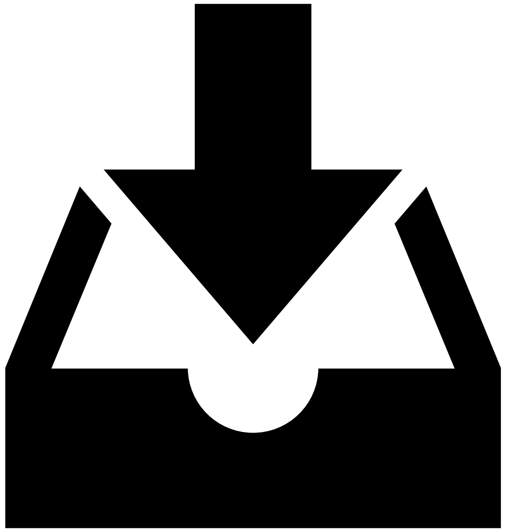
e.g. GeneA.Sample1 = $\frac{\frac{15}{2000}}{\left(\frac{15}{2000} + \frac{25}{4000} + \frac{10}{1000} + \frac{0}{10000}\right)} * 10^6 = 0.32 * 10^6\ TPM$

## EXTRA EXCERCISES

**Task 08. Using the compressed Fastqc file ABR3_D_03.fastq.gz (folder 01_RNAseq_raw), perform the steps of generating the QC report (FastQC software), filtering (Trimmomatic software) the sequences to remove adapters and low quality (and QC report after filtering) and aligning (Kallisto software) the filtered sequences against the reference transcriptome.**

**Task 09. Using Sleuth and the same input as in the previous example, visualize and compare the expression of the Bradi3g51200.1, Bradi1g37410.1, Bradi3g43870.1 transcripts using a heatmap.**
**What differences do you observe when comparing their expression in the control group versus under drought conditions?**

**Task 10. Construct a co-expression network (WGCNA R package) of the samples and genes analyzed under the control condition (W). Compare the results of the modules obtained in both co-expression networks, drought (D) and control (W).**



# Deliverable Task

Please, make a folder named ‘transcripts/’ in the same GitHub repo of session 1, and write a report that includes:

- ✓ How you solved the exercises.
- ✓ The results obtained.
- ✓ The main specific inputs and outputs used at each step.
- ✓ The problems you encountered.

Briefly propose an experiment in which these analyses could be applied.

# Thank you!

**If you have any questions, please feel free to email me at**

rsancho@eead.csic.es